Cloud Native Data Engineering

Week 3. Relational Databases

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Lecture Outline

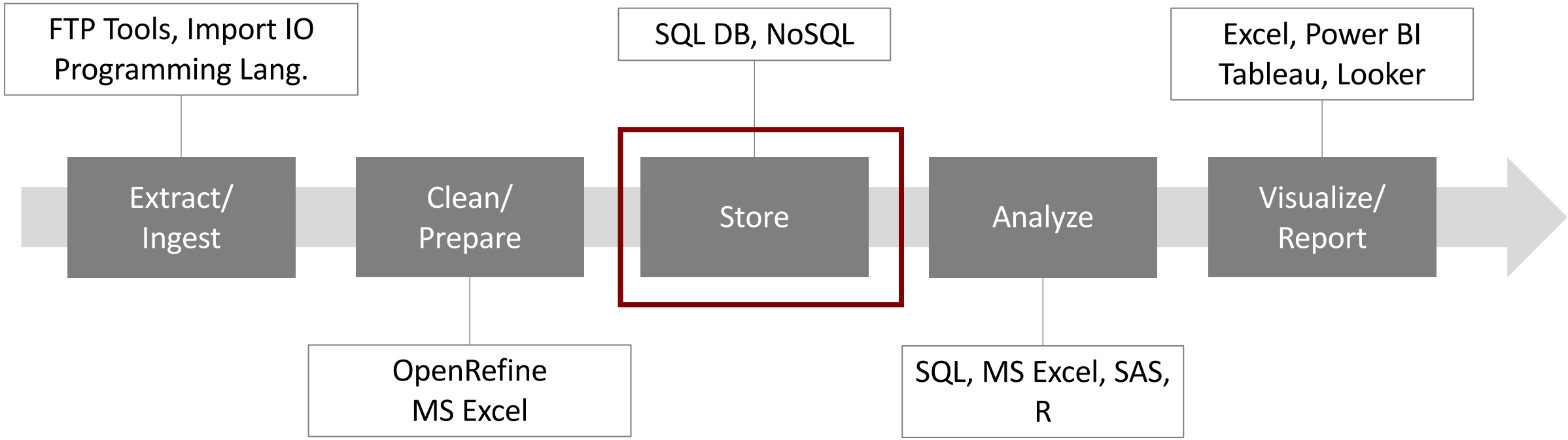
1. Databases and the DBMS
2. Conceptual Modeling
3. Logical Design: From ER to Tables
4. Normalization
5. Physical Design and Operations
6. ClassicModels Lab
7. Appendix: Reference, Exercises and Case Studies



Databases and the DBMS

Module 1

Data Storage and Analysis



Database

- Shared collection of logically related data and descriptions of this data
- Modeled after real-world systems and entities



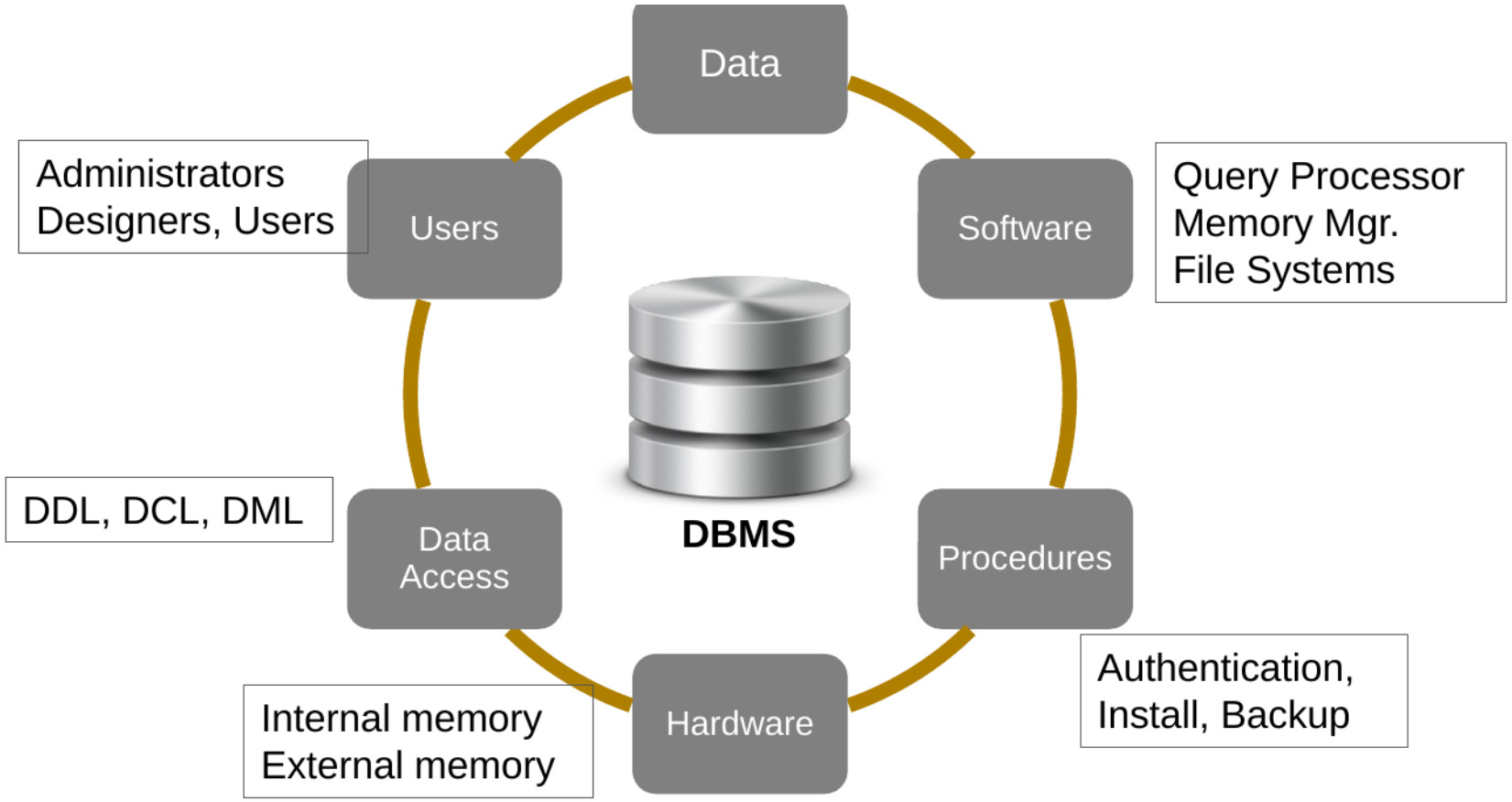
Database Management System (DBMS)

- Aggregate of data, software, hardware and users that helps an organization manage its operational data
- Provide efficient and reliable methods of data retrieval
- Involves monitoring, administration, and maintenance of the databases

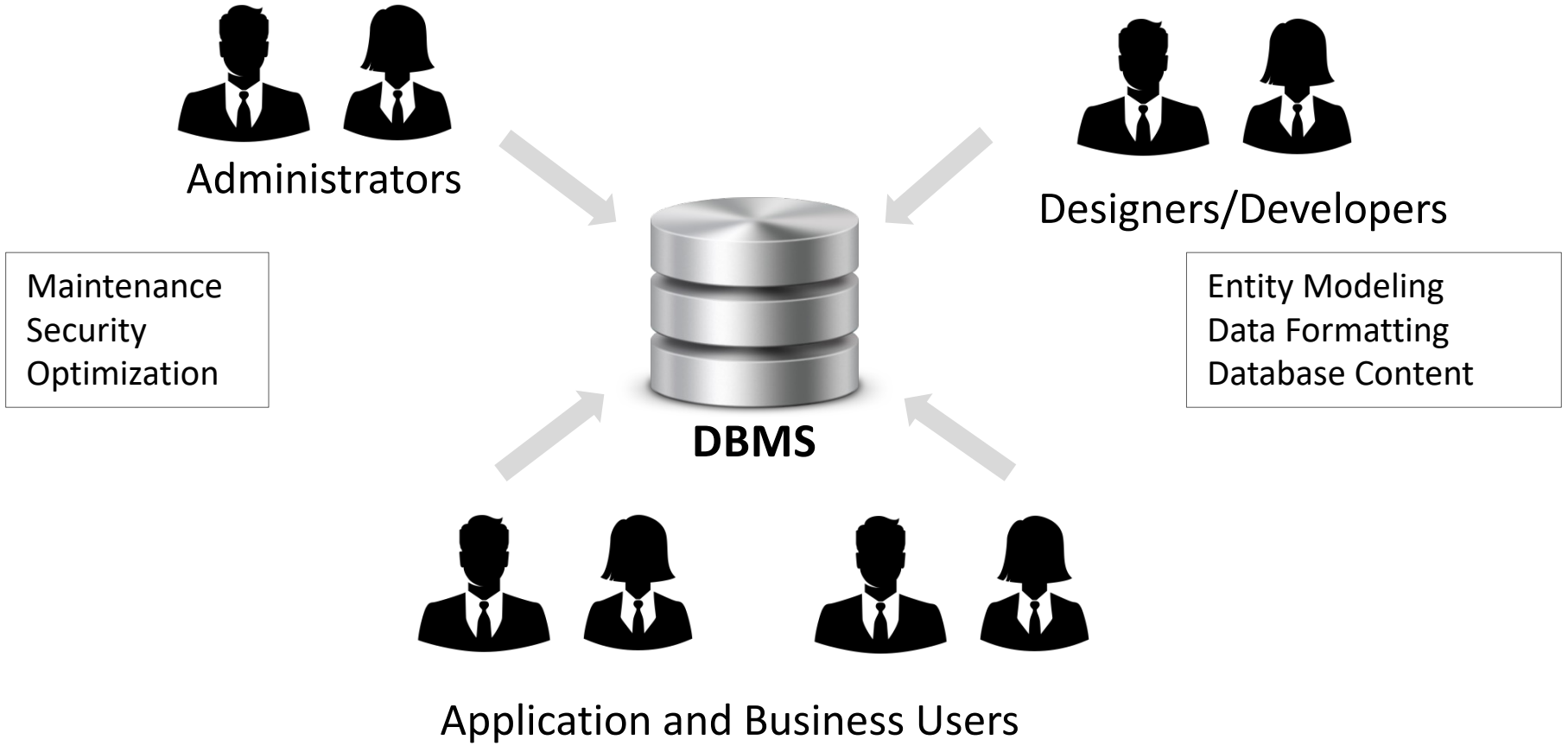
DBMS Benefits

- Program-Data Independence
- Efficiently retrieve, manipulate, store data
- Concurrent Access & Crash Recovery
- Minimize data duplication
- Data Integrity & Security
- Data Administration

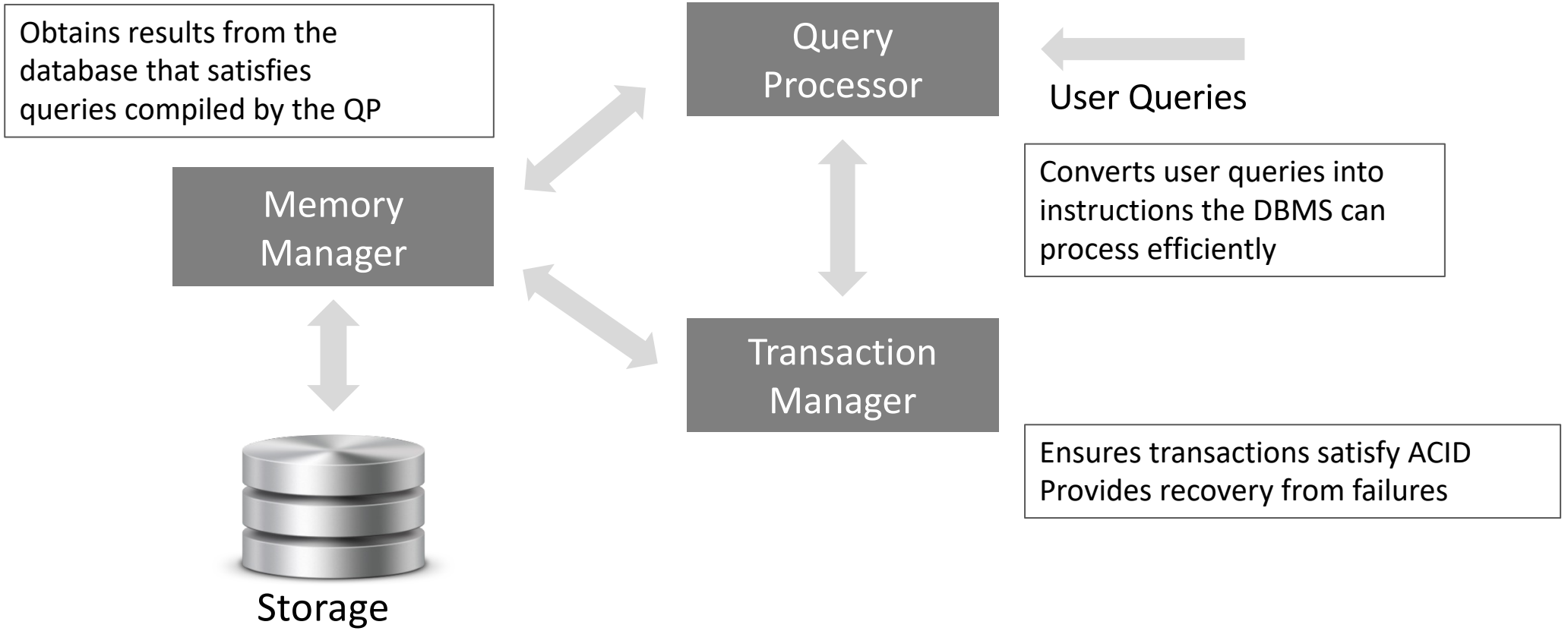
DBMS Components



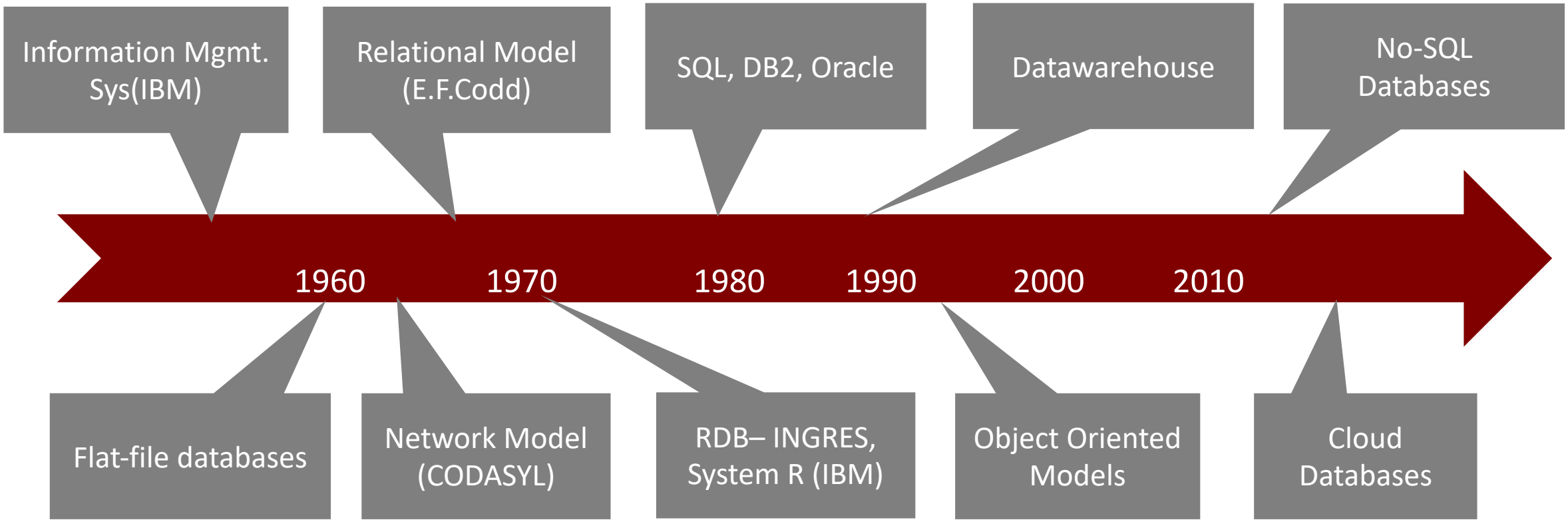
DBMS Users


Administrators
Designers/Developers
Maintenance
Security
Optimization
Entity Modeling
Data Formatting
Database Content
DBMS
Application and Business Users

DBMS Software Architecture

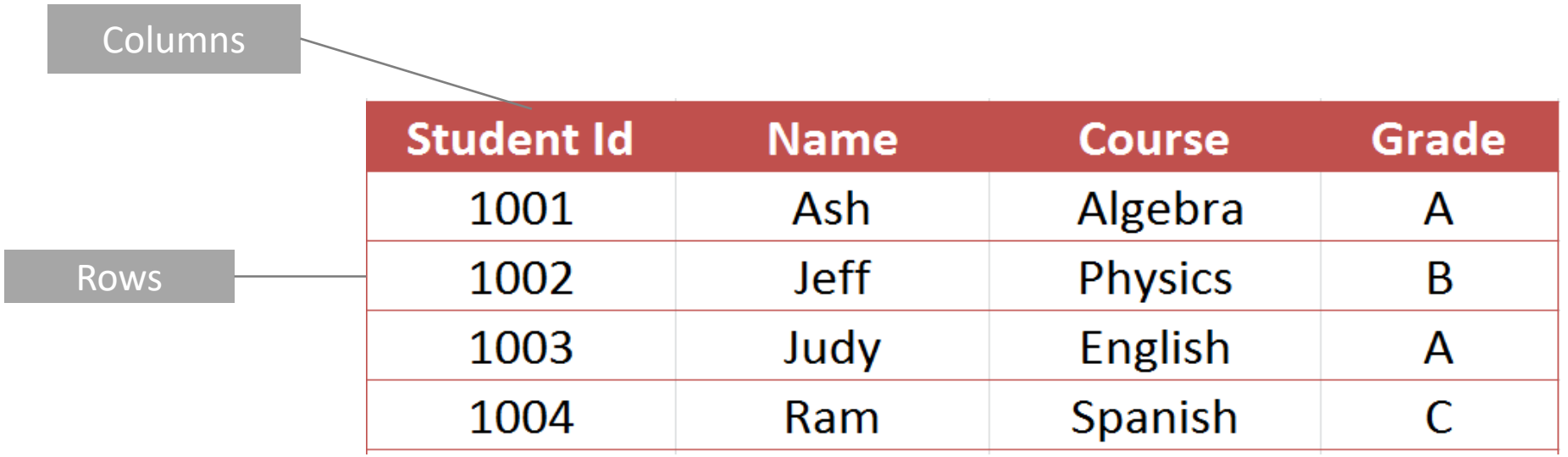


Evolution of Databases



Relational Databases

- Tables (relations) of rows (tuples) and columns (attributes), related through keys
- Codd's relational model (1970); queried with SQL; schema-on-write
- Examples: PostgreSQL, MySQL, SQL Server, Oracle, Cloud Spanner



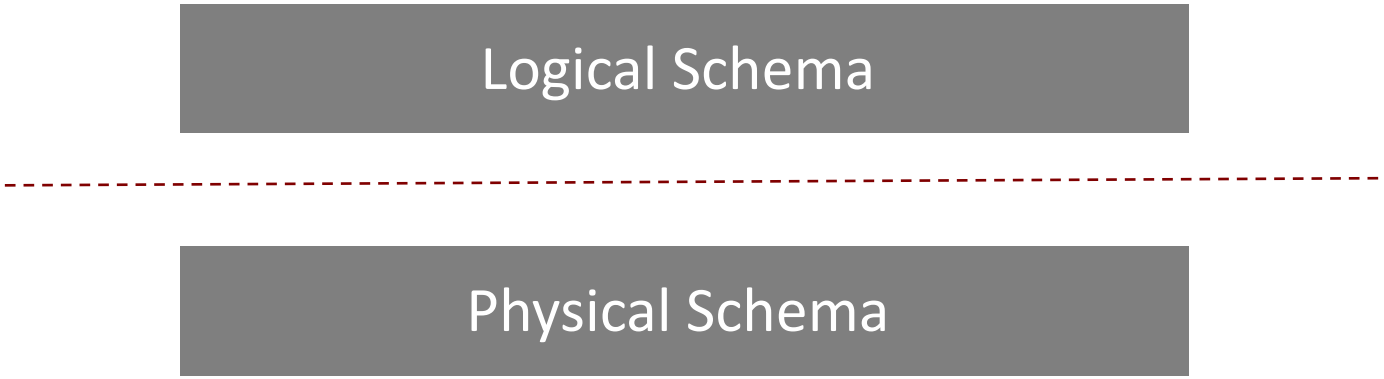


Conceptual Modeling

Module 2

Data Model and Schema

- Data model: abstract model that organizes data elements and their relationships
  - Data structure, data integrity, data manipulation
- Schema: the physical implementation of a data model in a specific DBMS
  - Tables, columns, keys, indexes (or graphs, hierarchies, objects in other models)
- Principle of Data Independence



Data model considerations

- What data elements will be stored?
- How large can each element be?
- What kind of information can each element contain?
- Which elements may be left blank?
- Which elements are constrained to a fixed range?
- How are the tables linked?

Database Design Process

1. Requirements analysis: gather business needs, questions, and constraints with stakeholders
2. Conceptual model: identify entities and relationships (ER diagram, tool-agnostic)
3. Logical model: map the ER model to tables, keys, and constraints; apply normalization
4. Physical model: choose data types, indexes, partitioning, and storage engine
5. Iterate with stakeholders; a good model saves years of migration pain

Conceptual, Logical, and Physical Models

CONCEPTUAL MODEL

- Model entities and relationships
- High level look at database
- Ignores hardware and software

LOGICAL MODEL

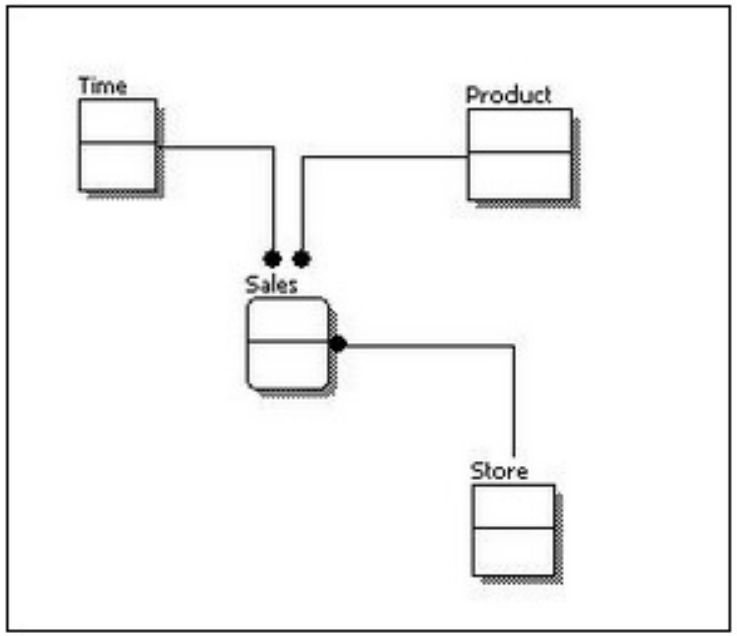
- Maps ER Model into relational database design
- Normalization of data
- Software specific
- Ignores hardware

PHYSICAL MODEL

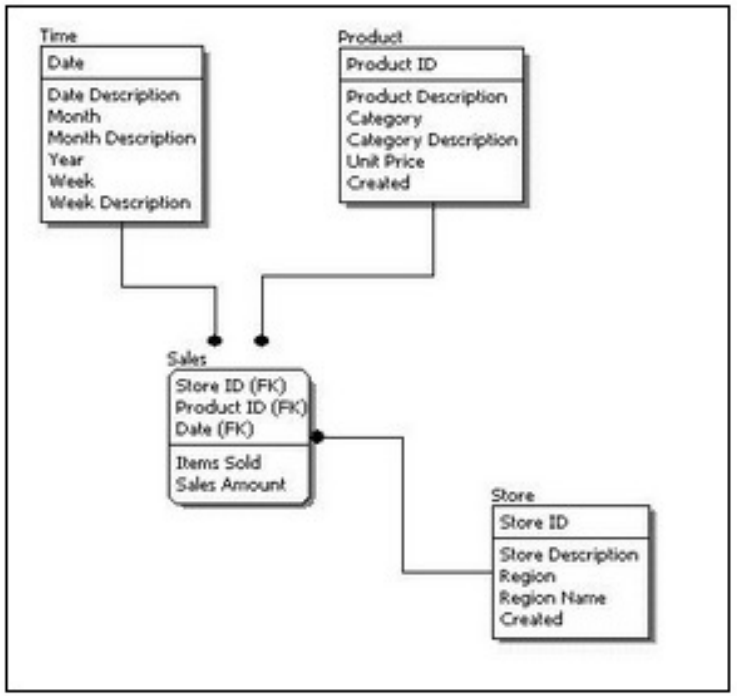
- Implements physical relational database tables
- Hardware and software dependent
- File structure and storage requirements

Three Levels of Model Design: Examples

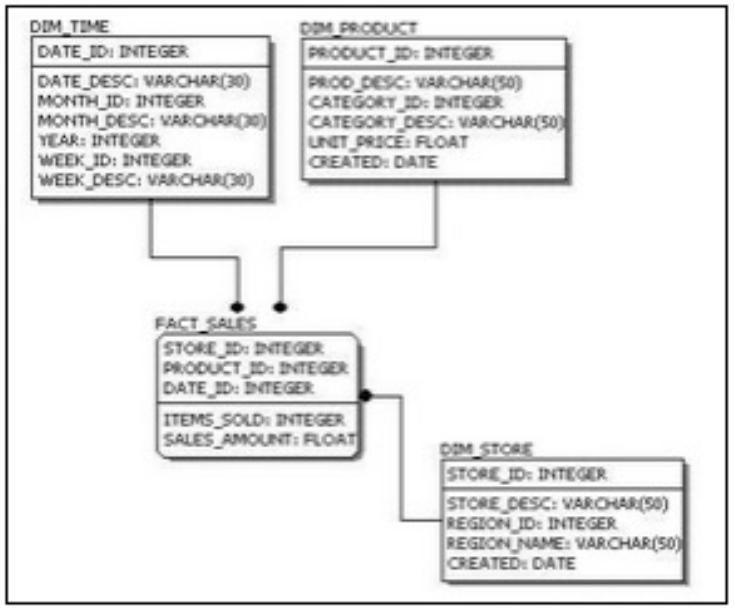
Conceptual Model Design



Logical Model Design



Physical Model Design



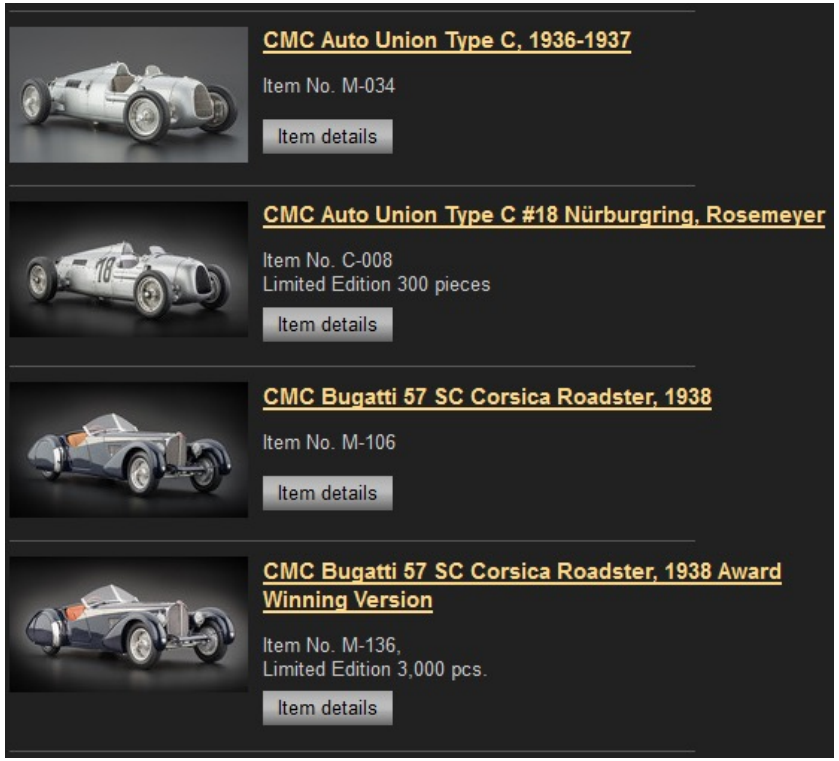
Source: www.1keydata.com

Conceptual vs. Logical vs. Physical Models

Feature	Conceptual	Logical	Physical
Entity Names	✓	✓	✓
Entity Relationships	✓	✓	✓
Attributes		✓	✓
Primary Keys		✓	✓
Foreign Keys		✓	✓
Table Names			✓
Column Names			✓
Column Data Types			✓

Case Study: ClassicModels

- Company selling classic model cars
- Objectives are to understand :
  - Sales of different products
  - Customers purchasing these products
  - Effectiveness of the sales team, etc.



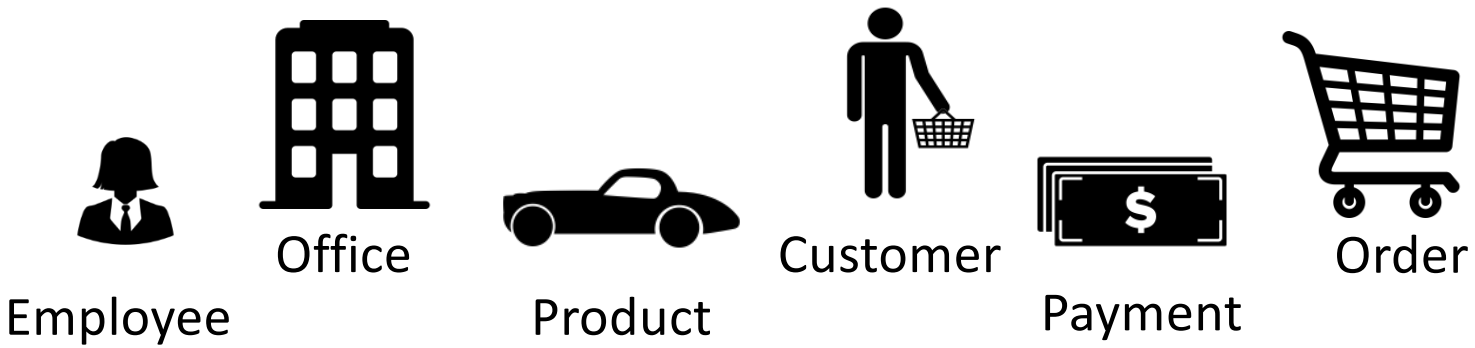
Img source: www.cmc-modelcars.de

Entity

- Real world object such as a person, place, thing or an abstract object such as a course, a flight reservation, etc.
- Entity set is a collection of similar types of entities and is represented by a table
- Entity is represented by a solid box as shown here



Entity



Entity Identification Steps

1. Gather and examine the business requirements
2. Examine the nouns: are they items of significance?
3. Name each entity
4. Is there information of interest that the business needs to hold?
5. Is each instance of the entity uniquely identifiable?
6. Which attribute or attributes could serve as its unique identifier?
7. Write a description of the entity
8. Diagram each entity and a few of its attributes

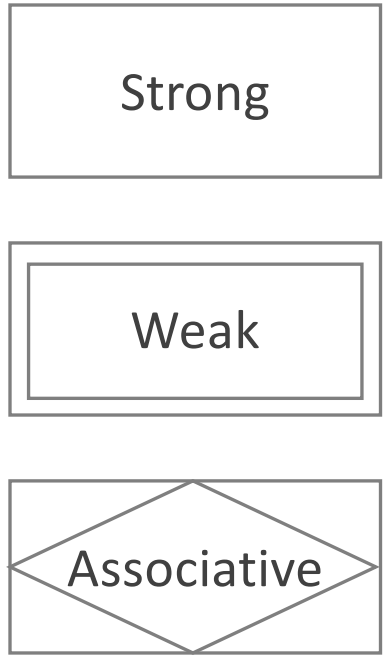
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Entity Descriptions

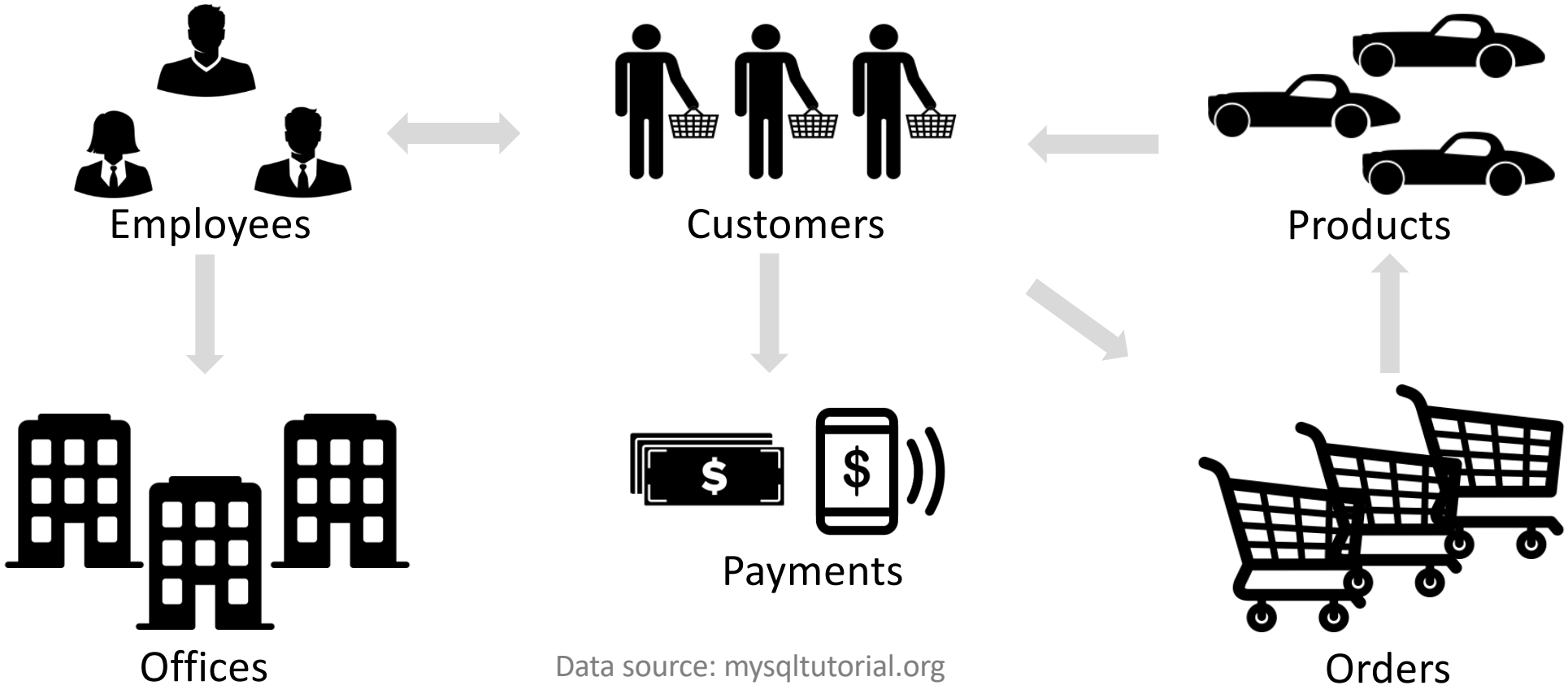
Entity	What it stores
Customers	Customer data
Products	List of scale model cars
ProductLines	Product line categories
Orders	Sales orders placed by customers
OrderDetails	Line items for each sales order
Payments	Payments made by customers against their accounts
Employees	Employee information and the organization structure (who reports to whom)
Offices	Sales office data

Entity Types

- Strong entities - exist independently from other entity types
- Weak entities - depend on some other entity type
- Associative entities - associate the instances of one or more entity types



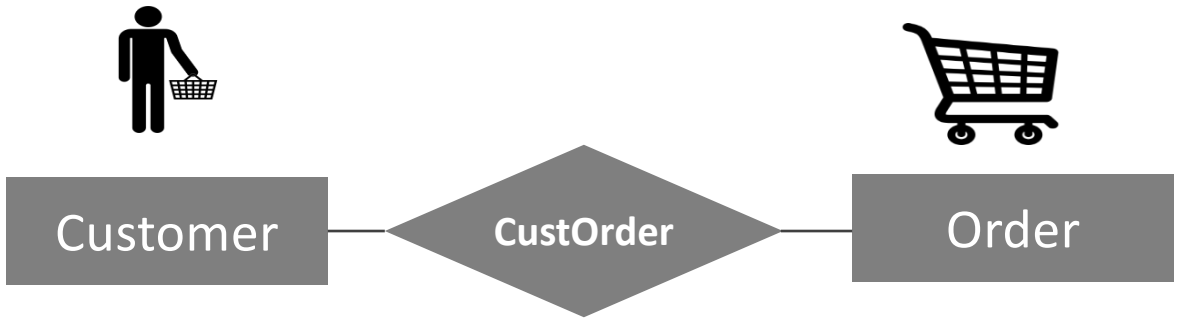
ClassicModels Entities



Data source: mysqltutorial.org

Relationship

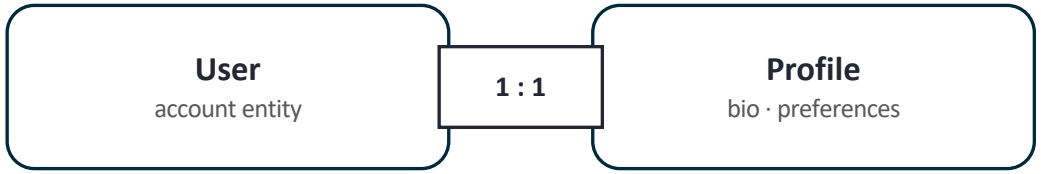
- Interactions or association between entities
- Degree of Relationship
  - Binary, Unary and Ternary
- Cardinality
  - Number of entities in one entity set, which can be associated with the number of entities of other set
  - One-to-One (1:1)
  - One-to-Many (1:M)
  - Many-to-Many (M:N)



Entity Relationship

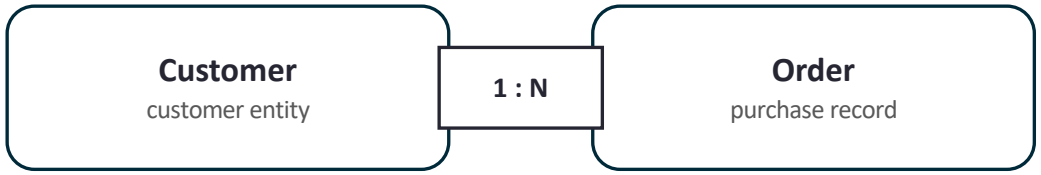
Relationships and Cardinality

ONE-TO-ONE



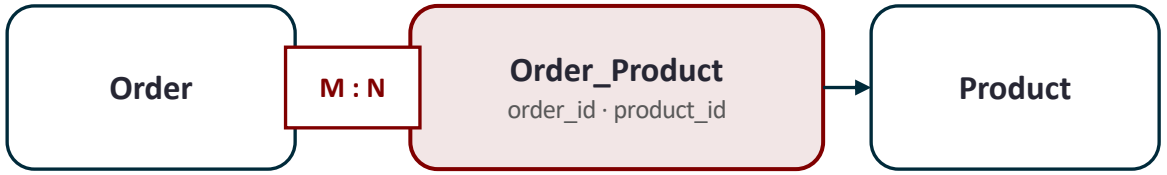
Each user has exactly one profile

ONE-TO-MANY



One customer places many orders

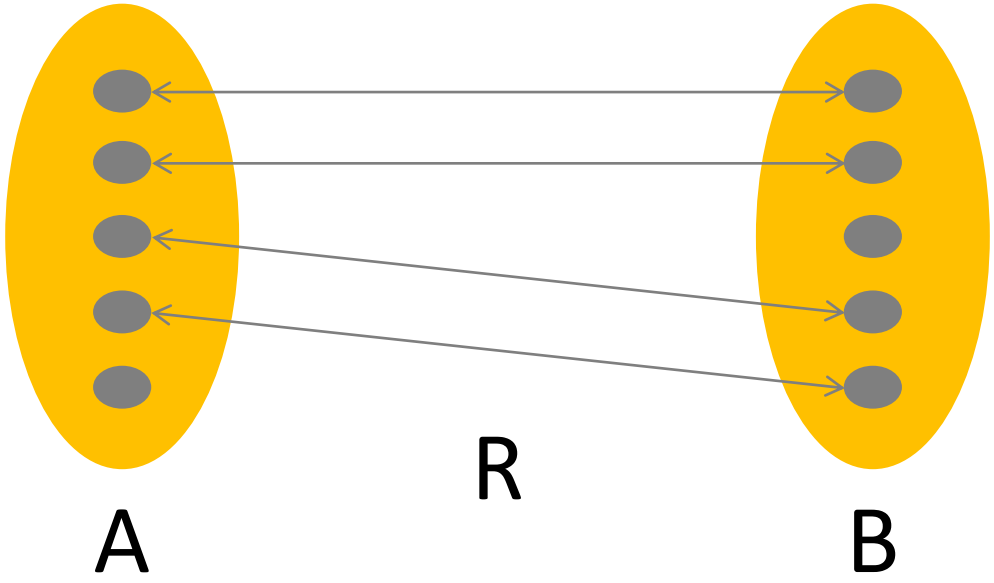
MANY-TO-MANY

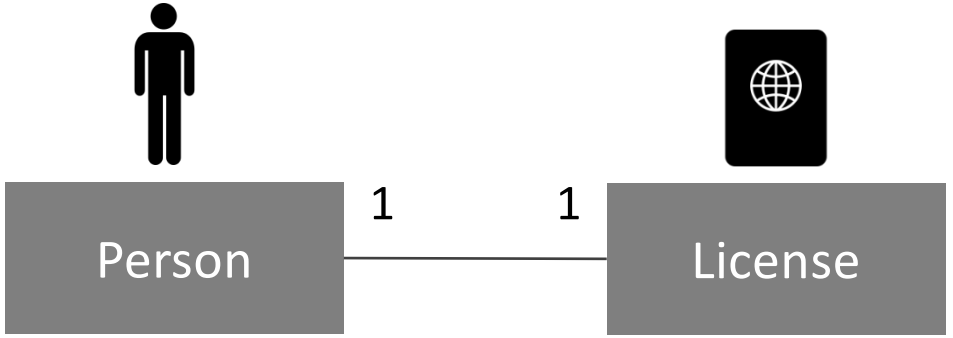


Many orders contain many products resolved via a junction table

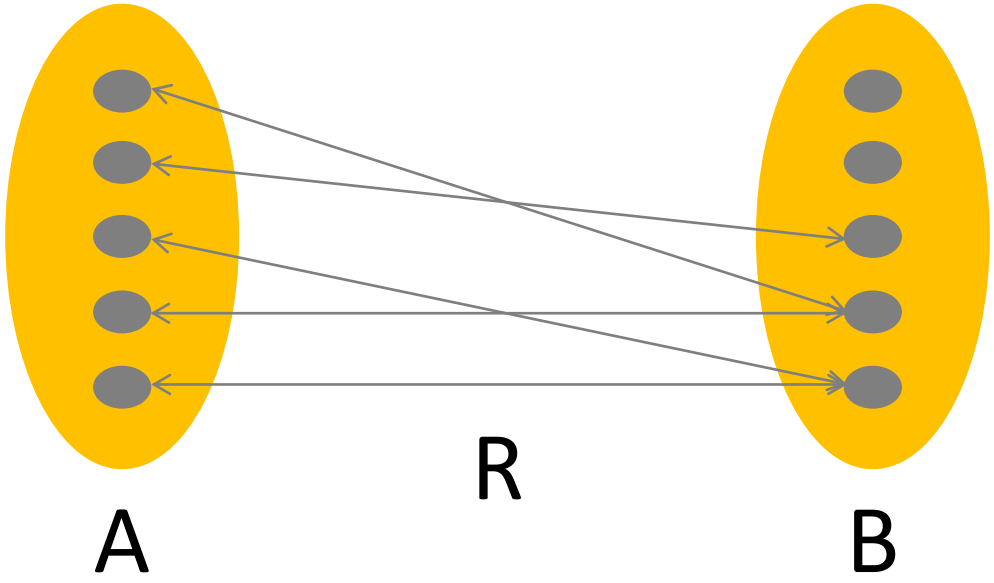
Junction tables decompose M:N into two 1:N relationships — each with a foreign key reference

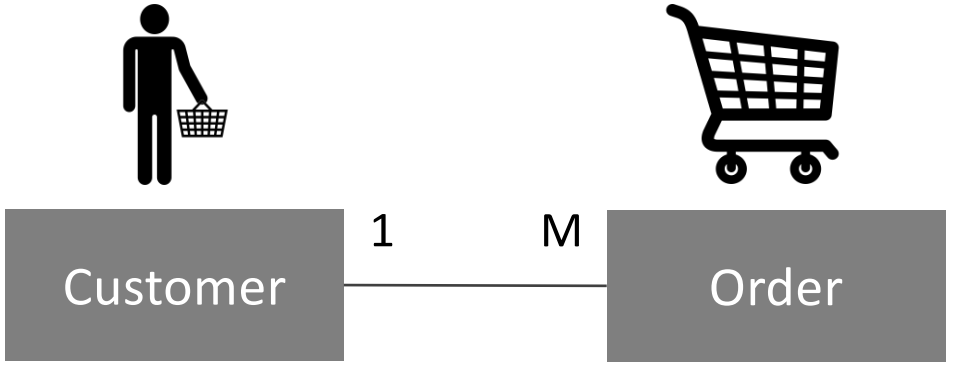
One-to-One (1:1)



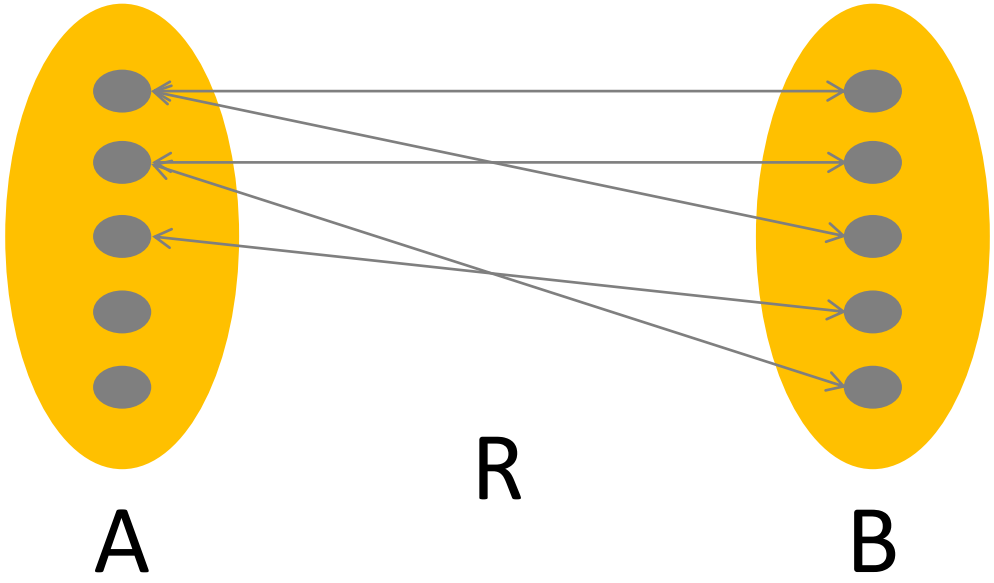


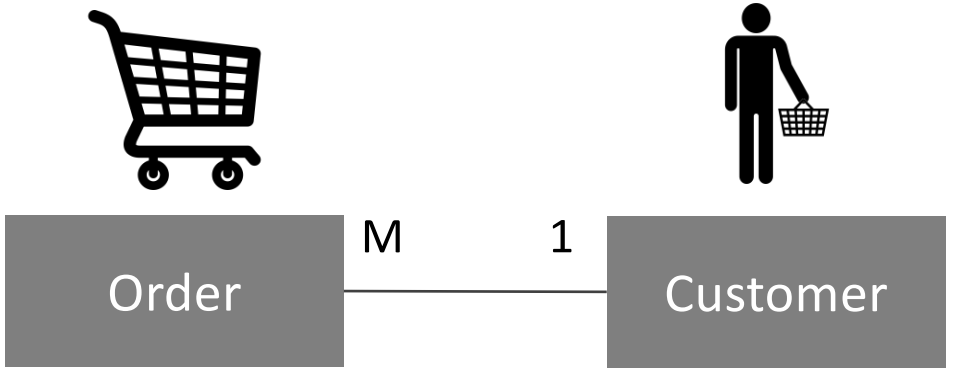
One-to-Many (1:M)



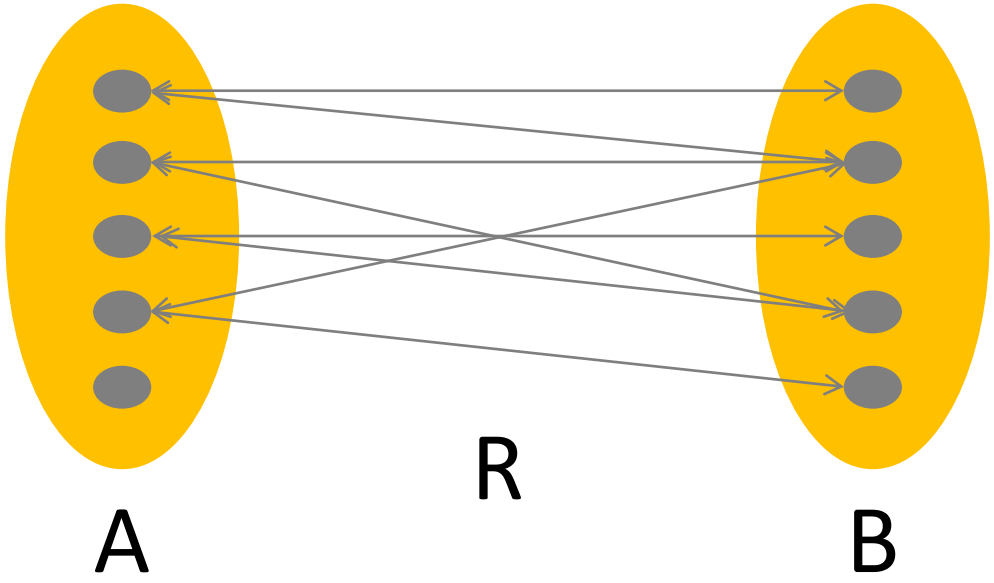


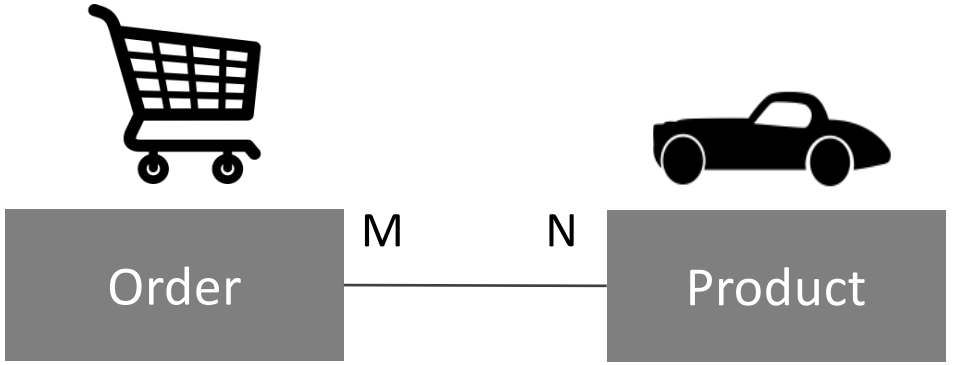
Many-to-One (M:1)





Many-to-Many (M:N)





Cardinality – ER Notation



One



One (and only one)



Zero or One



Many



One or Many



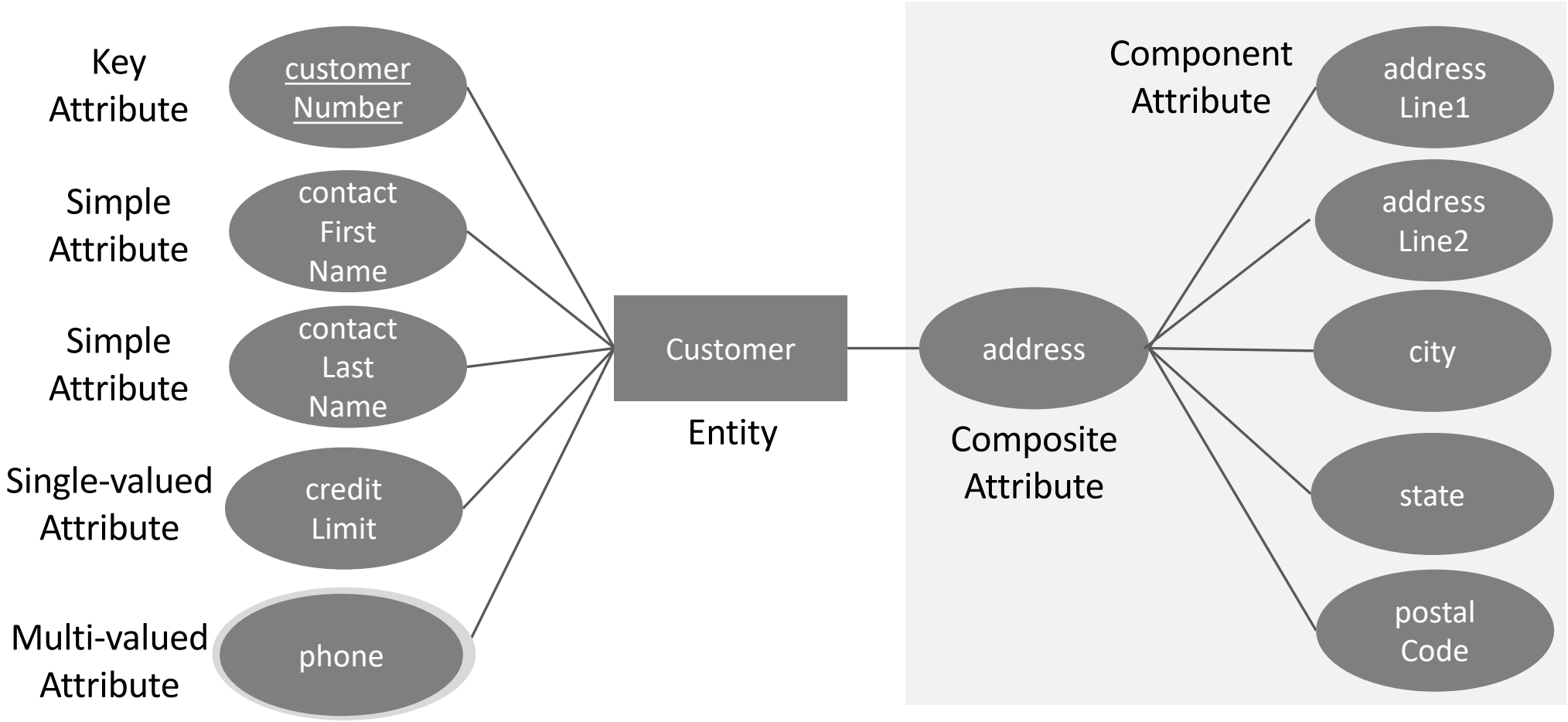
Zero or Many

Relationship Modeling Steps

1. Identify the first entity
2. Identify the related entity
3. Identify the optionality (must be or may be)
4. Identify the relationship
5. Identify the cardinality (1:N, 0:N, M:N, etc.)

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Attribute Types



ClassicModels Attributes (1 of 2)



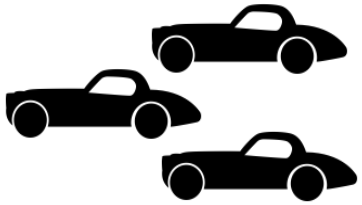
Employees

employeeNumber (numeric)
lastName (string)
firstName (string)
extension (string)
email (string)
officeCode (string)
reportsTo (string)
jobTitle (string)



Offices

officeCode (string)
city (string)
phone (string)
addressLine1 (string)
addressLine2 (string)
state (string)
country (string)
postalCode (string)
territory (string)



Products

productCode (string)
productName (string)
productLine (string)
productScale (string)
productVendor (string)
productDescription (string)
quantityInStock (numeric)
buyPrice (numeric)
MSRP (numeric)

ClassicModels Attributes (2 of 2)



Customers

customerNumber (string)
customerName (string)
contactLastName (string)
contactFirstName (string)
phone (string)
addressLine1 (string)
addressLine2 (string)
city (string)
state (string)
postalCode (string)
country (string)
salesRepEmployeeNumber (string)
creditLimit (numeric)



Orders

orderNumber (numeric)
orderDate (date)
requiredDate (date)
shippedDate (date)
status (string)
comments (string)
customerNumber (numeric)



Payments

checkNumber(string)
paymentDate (date)
amount (numeric)
customerNumber (numeric)

ClassicModels EER Diagram





Logical Design: From ER to Tables

Module 3

Logical Model Design Steps

1. Convert entities into tables
2. Convert relationships into foreign keys
3. Convert attributes into columns
4. Modify the physical data model based on requirements

Entity-Relational Mapping



Employees

- employees
  - Columns
    - employeeNumber
    - lastName
    - firstName
    - extension
    - email
    - officeCode

employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
1002	Murphy	Diane	x5800	dmurphy@classicmodelcars.com	1	NULL	President
1056	Patterson	Mary	x4611	mpatterso@classicmodelcars.com	1	1002	VP Sales
1076	Firrelli	Jeff	x9273	jfirrelli@classicmodelcars.com	1	1002	VP Marketing
1088	Patterson	William	x4871	wpatterson@classicmodelcars.com	6	1056	Sales Manager (APAC)
1102	Bondur	Gerard	x5408	gbondur@classicmodelcars.com	4	1056	Sale Manager (EMEA)
1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1143	Sales Rep

Keys



- Provide the means for database users and applications to identify, access and update information in tables.
- A set of one or more attributes of a table
  - Simple Key
  - Compound Key
  - Primary Key (PK)
  - Foreign key (FK)

Primary Key (PK)

- An attribute (or combination of attributes) that uniquely identifies any given entity
- Primary key cannot be null
- Composite primary key
  - PK made up of multiple columns
- Candidate key
  - A minimal super key: no subset of its attributes is itself a super key

Information

Table: employees

Columns:

employeeNumber	int(11) PK
lastName	varchar(50)
firstName	varchar(50)
extension	varchar(10)
email	varchar(100)
officeCode	varchar(10)
reportsTo	int(11)
jobTitle	varchar(50)

Related Tables:

Target	employees (reportsTo → employeeNumber)
On Update	RESTRICT
On Delete	RESTRICT
Target	offices (officeCode → officeCode)
On Update	RESTRICT
On Delete	RESTRICT

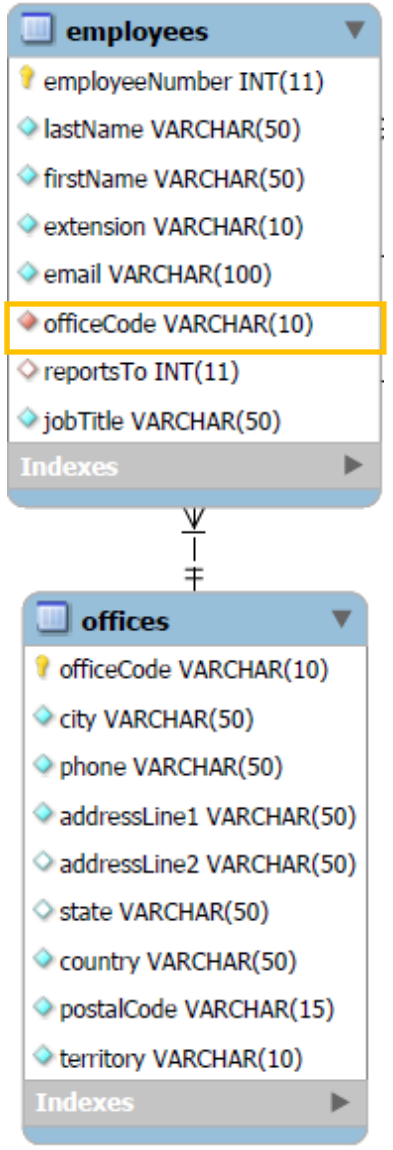
Natural vs Surrogate Keys

- Natural key: a real-world identifier (email, SSN, product code); meaningful, but can change
- Surrogate key: system-generated (auto-increment, UUID); stable, carries no business meaning
- Choose by asking: Is it stable? Is it minimal? Do users search by it? Does it meet performance needs?

Employee ID	SSN	Employee Name	Office
1	093-009-1385	Sarah	Boston
2	026-989-2900	Wei	Chicago
3	823-111-2270	Ram	New York

Foreign Key (FK)

- Primary key from one table placed into another table
- Used to create relationships between entities
- Every value of a FK must match a value of an existing PK

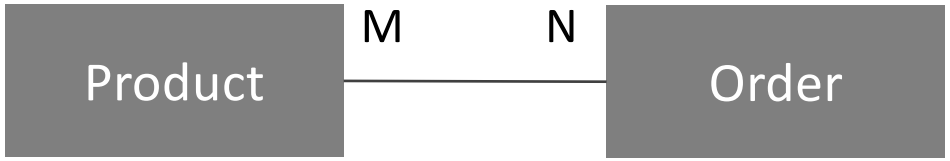


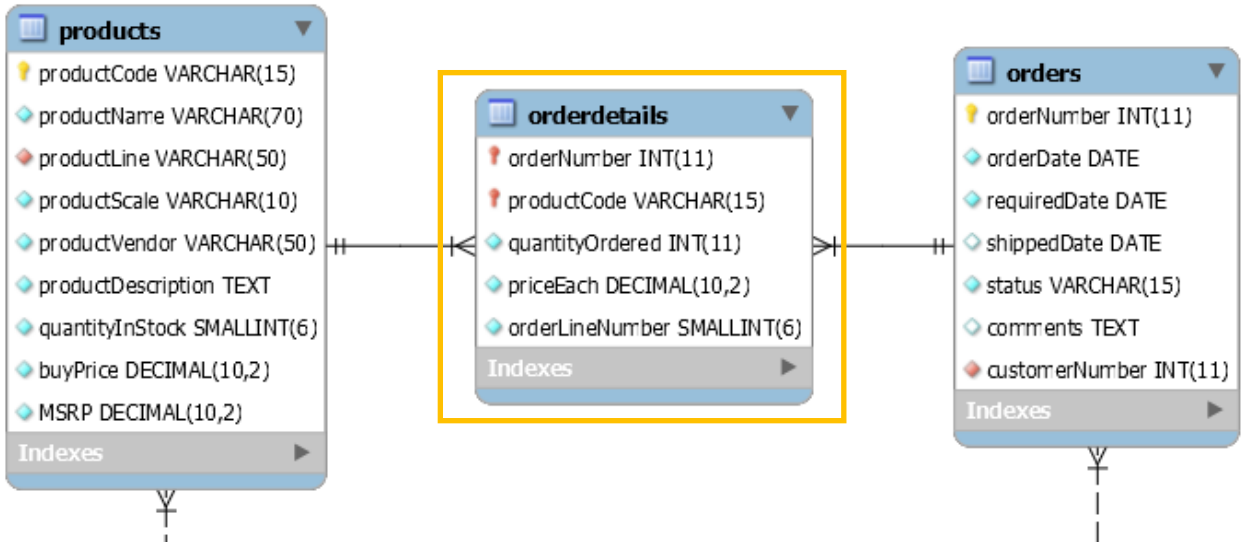
Constraints and Data Integrity

Integrity type	Rule	ClassicModels example
Entity integrity	PRIMARY KEY and UNIQUE: no duplicate or NULL keys	employeeNumber cannot be null or empty
Referential integrity	FOREIGN KEY values must match an existing PK: no orphan rows	A payment must link to a customer
Domain integrity	Data types, NOT NULL, CHECK, DEFAULT	$100 < creditLimit < $1,000,000

Join Table

- Used to model many-to-many relationships
  - Implemented as two 1:N relationships to a junction table





Lookup Tables

- Tables containing reference or lookup data
- Result of normalization process
- Improve efficiency in OLTP
- Too many lookups during joins impact OLAP performance

City (PK)	State	Country
San Francisco	CA	USA
Boston	MA	USA
NYC	NY	USA

Status ID	Status Description
1	Shipped
2	Cancelled
3	Resolved



Normalization

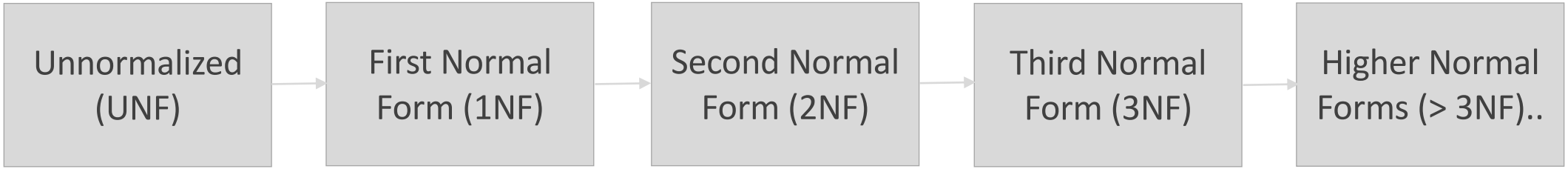
Module 4

Normalization

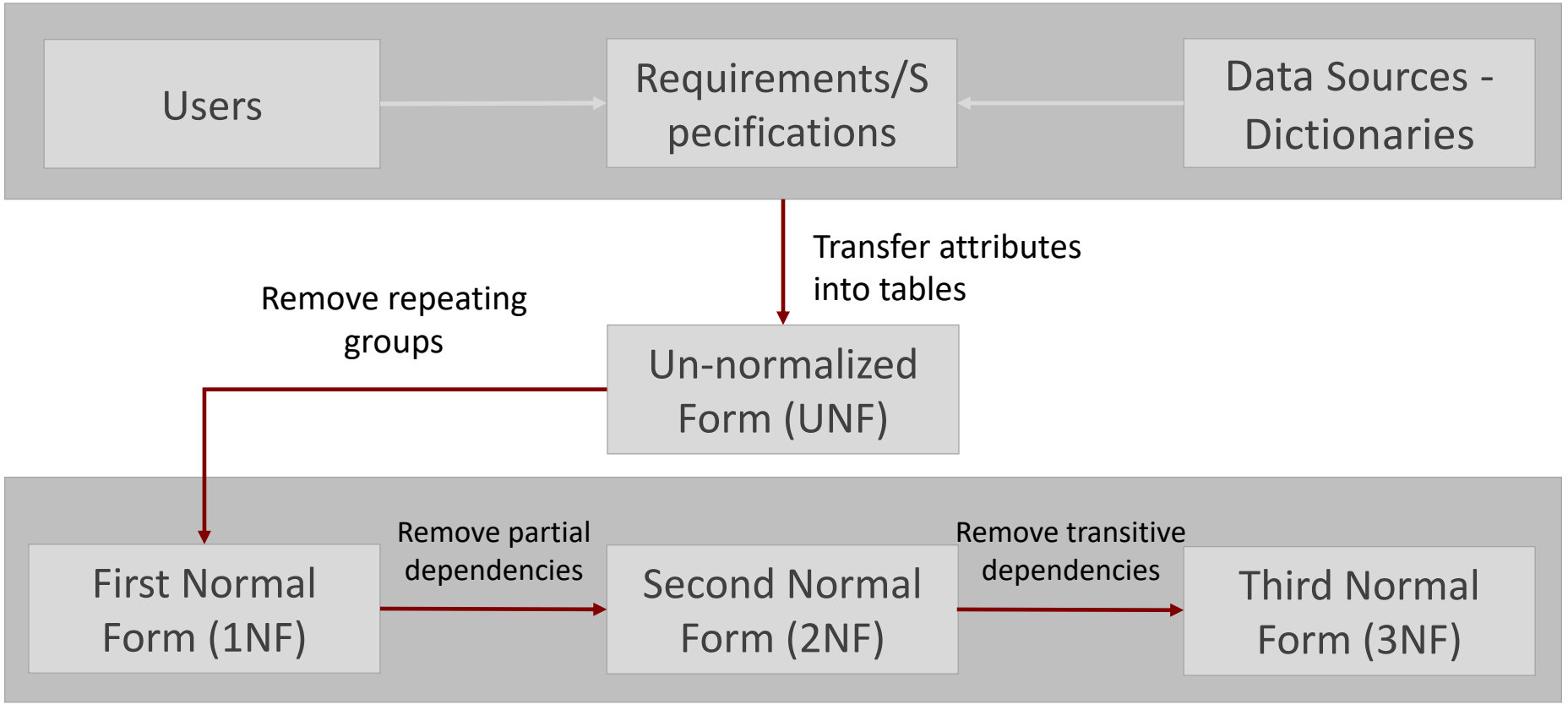
- Improves relational table design to minimize data redundancy
- Involves increasing the number of database tables
- Degrees of normalization:
  - 1NF, 2NF, 3NF, BCNF, 4NF, 5NF

Normalization Process

- Normalization is done in stages; each stage applies more rigorous rules
- Apply the normalization rules to each table:
  1. Examine the table and check whether it satisfies a particular normal form
  2. If yes, check whether it satisfies the next higher normal form
  3. If no, convert the table into a set of tables that do satisfy that normal form



Normalization Overview



Un-normalized Form and Redundancy Anomalies

- Un-normalized form (UNF): redundant data with repeating groups, non-atomic values, and no primary key
- Anomalies caused by redundancy:
  - Update: changing a value in one place leaves inconsistent copies elsewhere
  - Insert: some facts cannot be recorded without unrelated data
  - Delete: deleting a row unintentionally loses other facts

Office	Employee
Boston	Julie, Steve
Paris	Gerard, Pamela, Loui

1st Normal Form (1NF)

- Each row is unique; every column holds a single atomic value; no repeating groups
- UNF to 1NF: remove repeating groups by
  - flattening the table (one value per cell), or
  - moving the repeated data, with a copy of the original key, into a separate relation

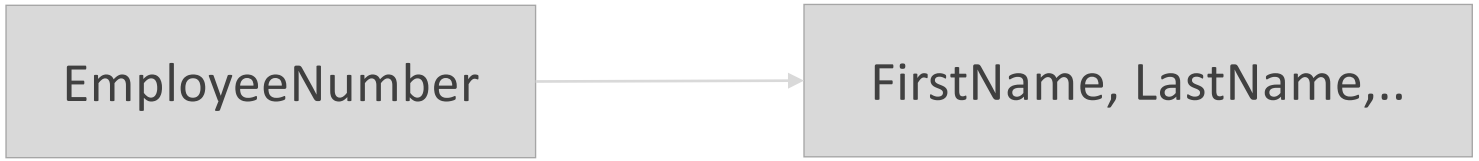
Office	Employee
Boston	Julie, Steve
Paris	Gerard, Pamela, Louis

Office	Employee
Boston	Julie
Boston	Steve
Paris	Gerard
Paris	Pamela
Paris	Louis

Functional Dependency

- Value of one (or more) column(s) in each record of a relation uniquely determines the value of another column

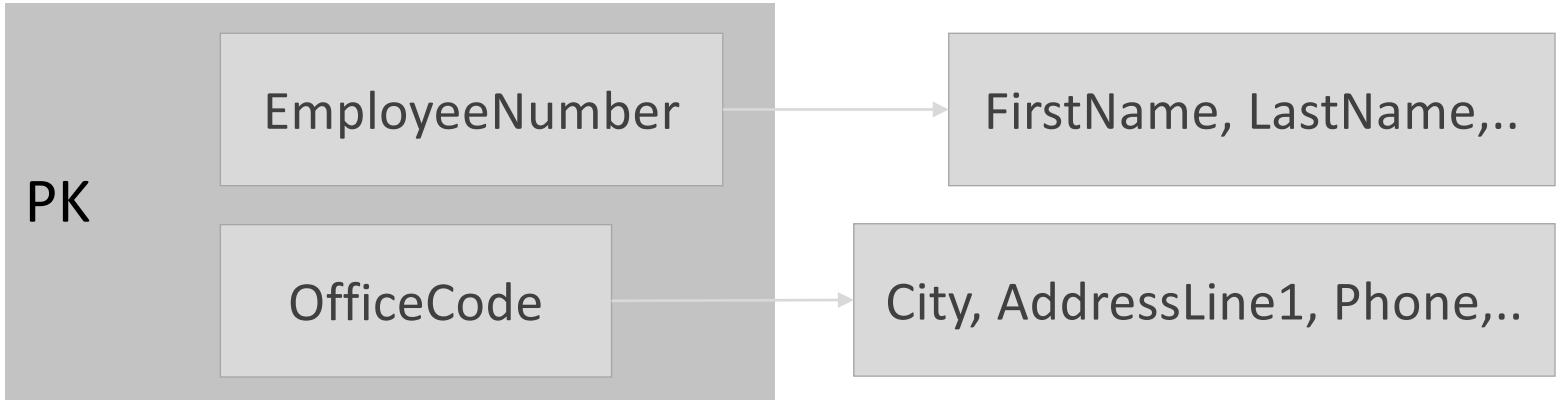
A and B are attributes of a relation. B is fully dependent on A if B is functionally dependent on A (A → B) but not on any proper subset of A



Partial Functional Dependency

- Column of a relation is functionally dependent on a component of the composite key

e.g. Employees and Offices data in one table

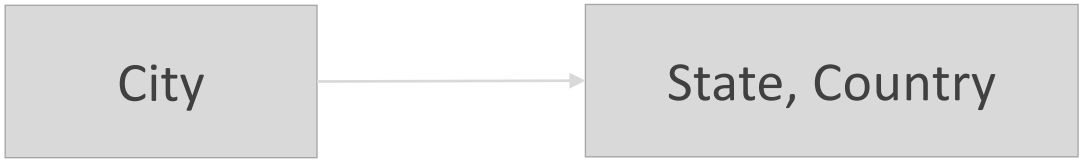


2nd Normal Form (2NF)

- In 1NF, and every non-key attribute depends on the whole primary key (no partial dependencies)
- To convert 1NF to 2NF:
  1. Identify the primary key and the functional dependencies
  2. Move partially dependent attributes to a new relation with a copy of their determinant

Transitive Functional Dependency

- Non-key columns functionally determine other non-key columns of a relation



3rd Normal Form (3NF)

- In 2NF, and non-key columns do not determine other non-key columns (no transitive dependencies)
- To convert 2NF to 3NF:
  1. Identify the primary key and the functional dependencies
  2. Move transitively dependent attributes to a new relation with a copy of their determinant

Normalization: Trade-offs

Advantages	Trade-offs
Less redundant data and storage; each table describes one entity	More tables and more joins; queries get more complex
Insert, update, and delete touch one place, so fewer anomalies	Codes replace descriptive values, so lookups are needed to read results
Smaller tables with primary-key indexes; join only what a query needs	Optimized for OLTP applications, not ad hoc analytics (warehouses often denormalize)



Physical Design and Operations

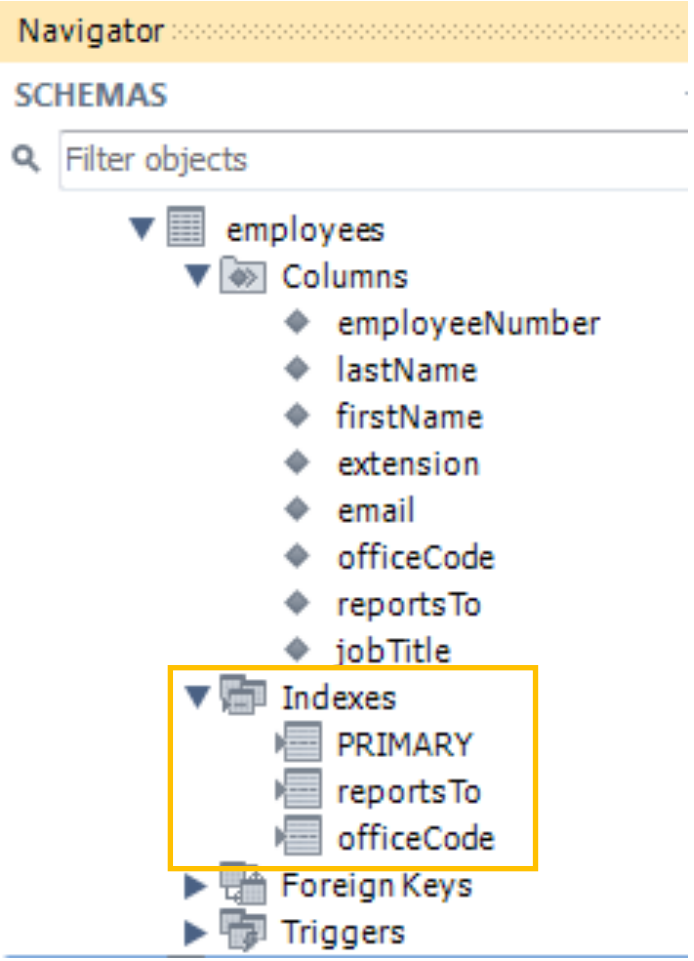
Module 5

Physical Design Considerations

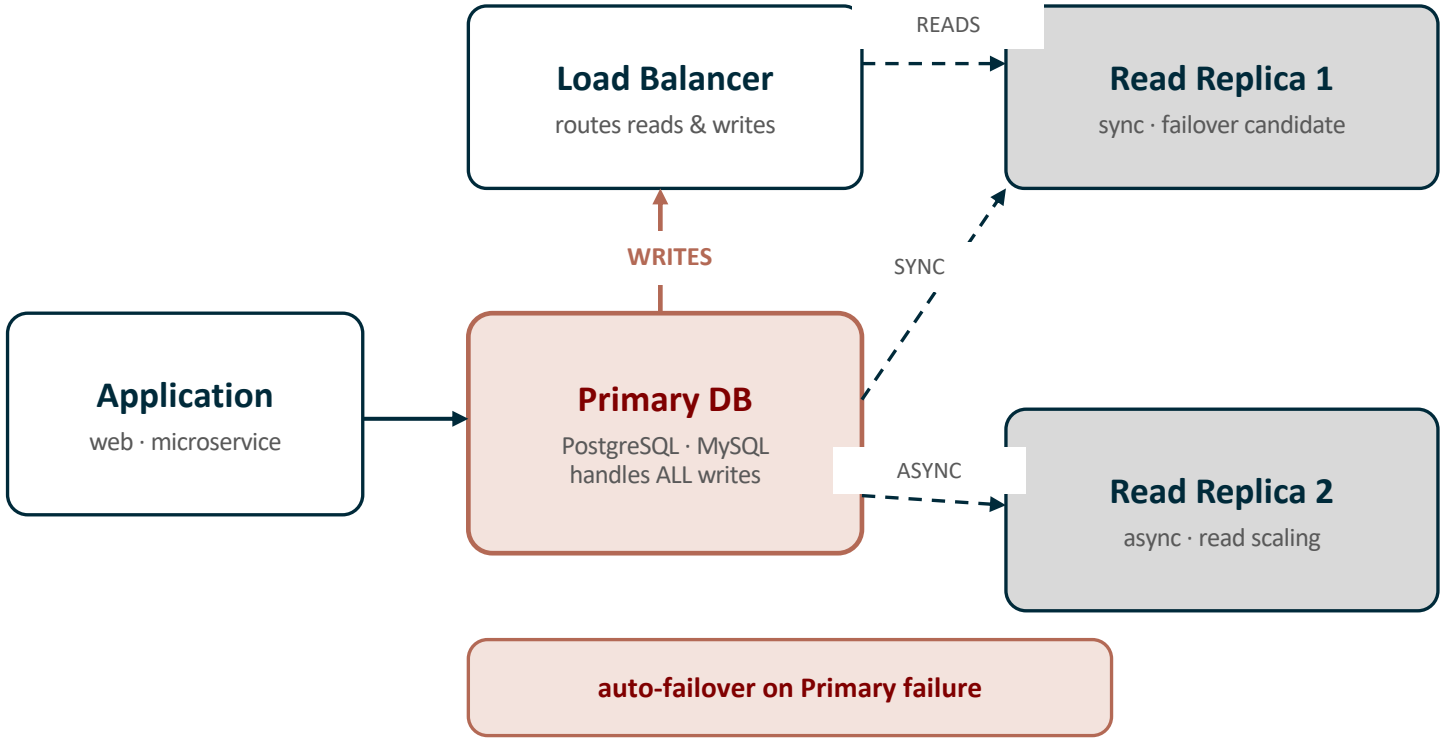
Technique	How it works	Best for
Range partitioning	Rows divided by value ranges of a key (date, region)	Time-series data; manageability of large tables
Hash partitioning	Rows distributed evenly by a hash of a key	Parallel loads and queries
Table clustering (CLUSTER BY)	Rows physically sorted by a column	Scan efficiency on large tables
Storage engine selection	InnoDB (ACID), MyISAM (speed), columnar (analytics)	Matching the engine to the workload

Indexing

- Data structure that retrieves records efficiently, built on chosen attributes; each index belongs to one table
- Clustered: stores row data in index order; efficient when data is sorted
- Non-clustered: separate structure, order differs from the physical data
  - Used for ORDER BY, WHERE, and JOIN



Replication and High Availability



TOPOLOGY

Primary-Replica

All writes → Primary
Reads scale horizontally

RPO / RTO

Sync: RPO ≈ 0
Async: higher throughput

Failover

Replica 1 promotes automatically on Primary failure

Writes always route to Primary · Reads distribute across replicas · Replica 1 promotes automatically on Primary failure

Backup and Recovery

Method	What it does	Notes
Full backup	Complete snapshot of the database; large	Taken weekly or daily
Incremental backup	Only changes since the last backup; smaller	Taken hourly
Point-in-time recovery (PITR)	Replays WAL/binlog from a backup to any moment in time	Managed backups with PITR are standard in Cloud SQL and RDS
Restore tests	Prove the backups actually restore	Test regularly: an untested backup may be corrupt

Database Security

Area	Practice
Access control	Application-specific database users; never connect as root or superuser; grant minimum privileges (SELECT only for reporting users)
Encryption in transit	Require SSL/TLS for all database connections
Encryption at rest	Full-disk or column-level encryption for PII
Auditing	Capture who changed what data and when on sensitive tables (audit all DML)

Monitoring Database Health

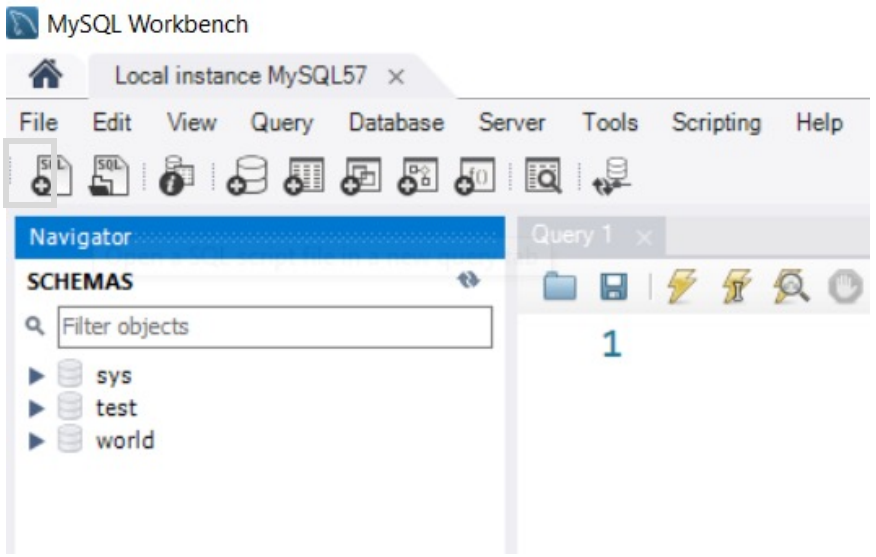
Signal	What it tells you	Example alert
Query latency (p50/p95/p99)	Overall responsiveness	p99 above the SLA
Connections used	Connection pool saturation	Above 80% of pool
Replication lag	How stale replicas are	Above 60 s
Slow query log	Queries over a threshold (analyze with pt-query-digest)	New slow queries after a release
Lock waits	Transactions holding locks too long	Growing wait times
Disk I/O saturation	Missing indexes or too small a buffer pool	Sustained high I/O wait

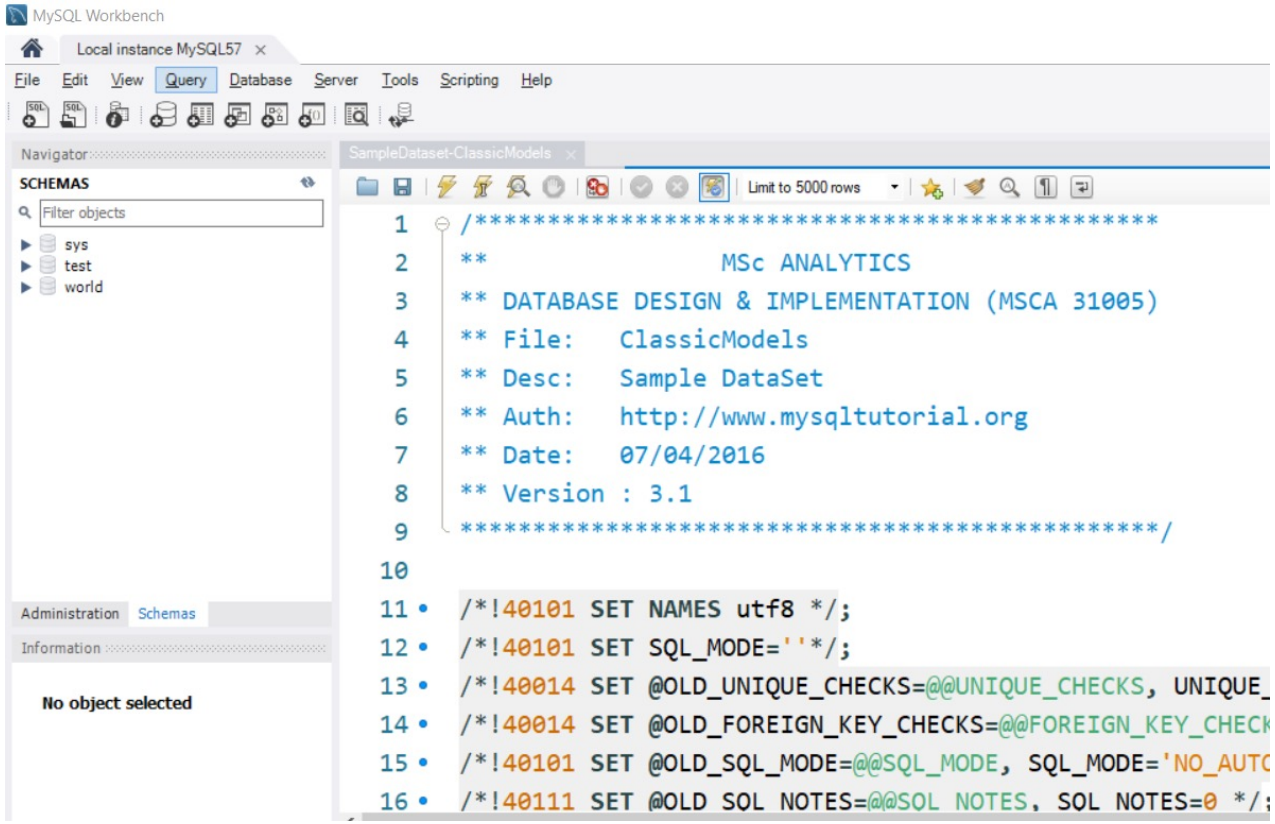


ClassicModels Lab

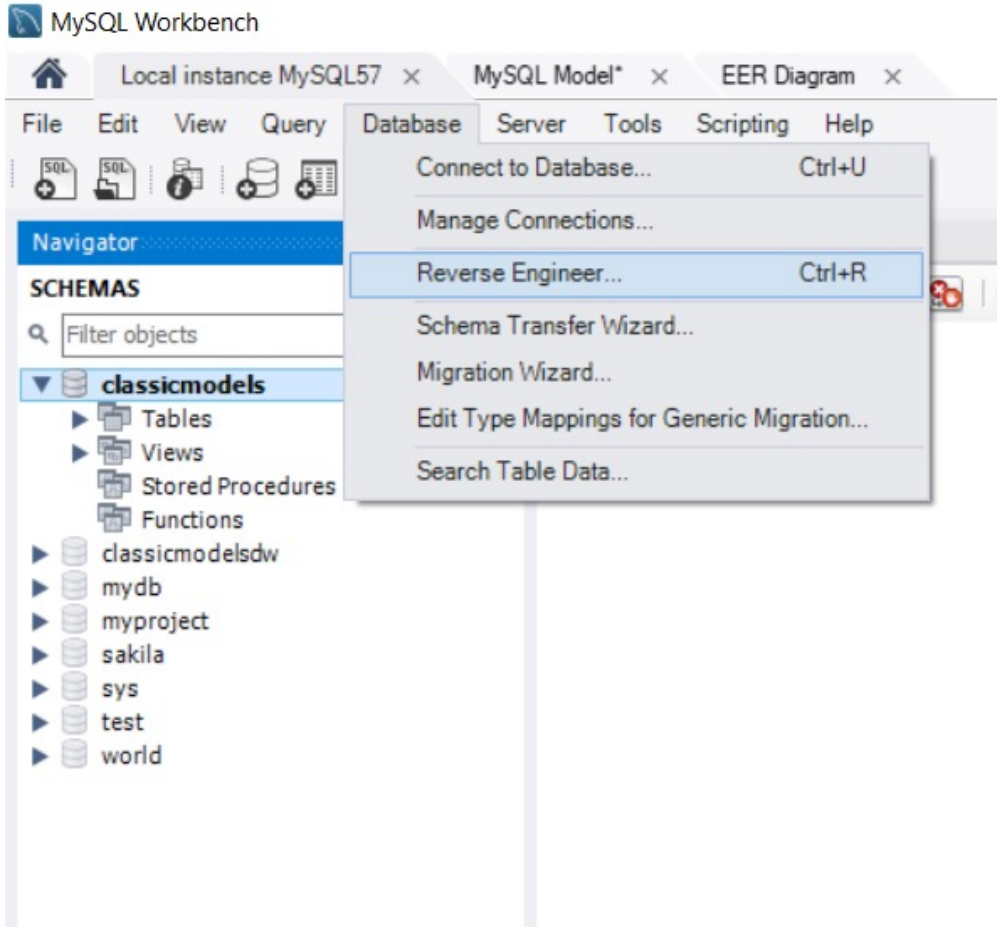
Module 6

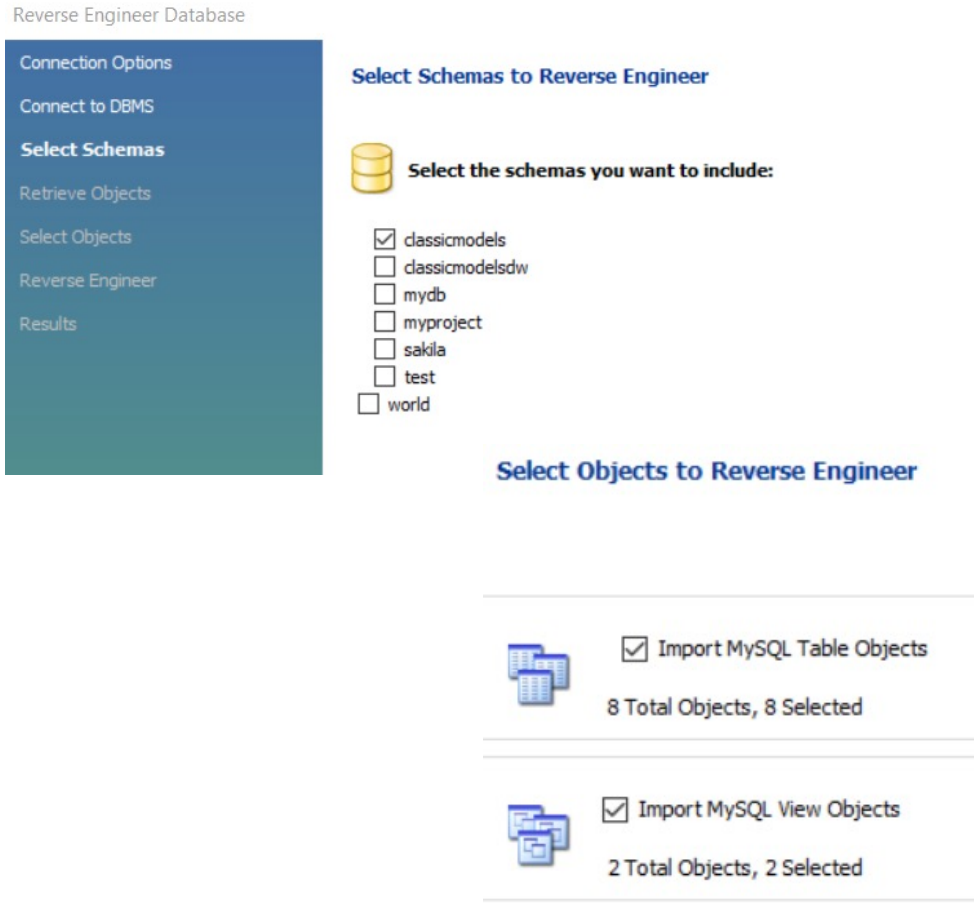
Lab 1: Create the Database Using a SQL Script



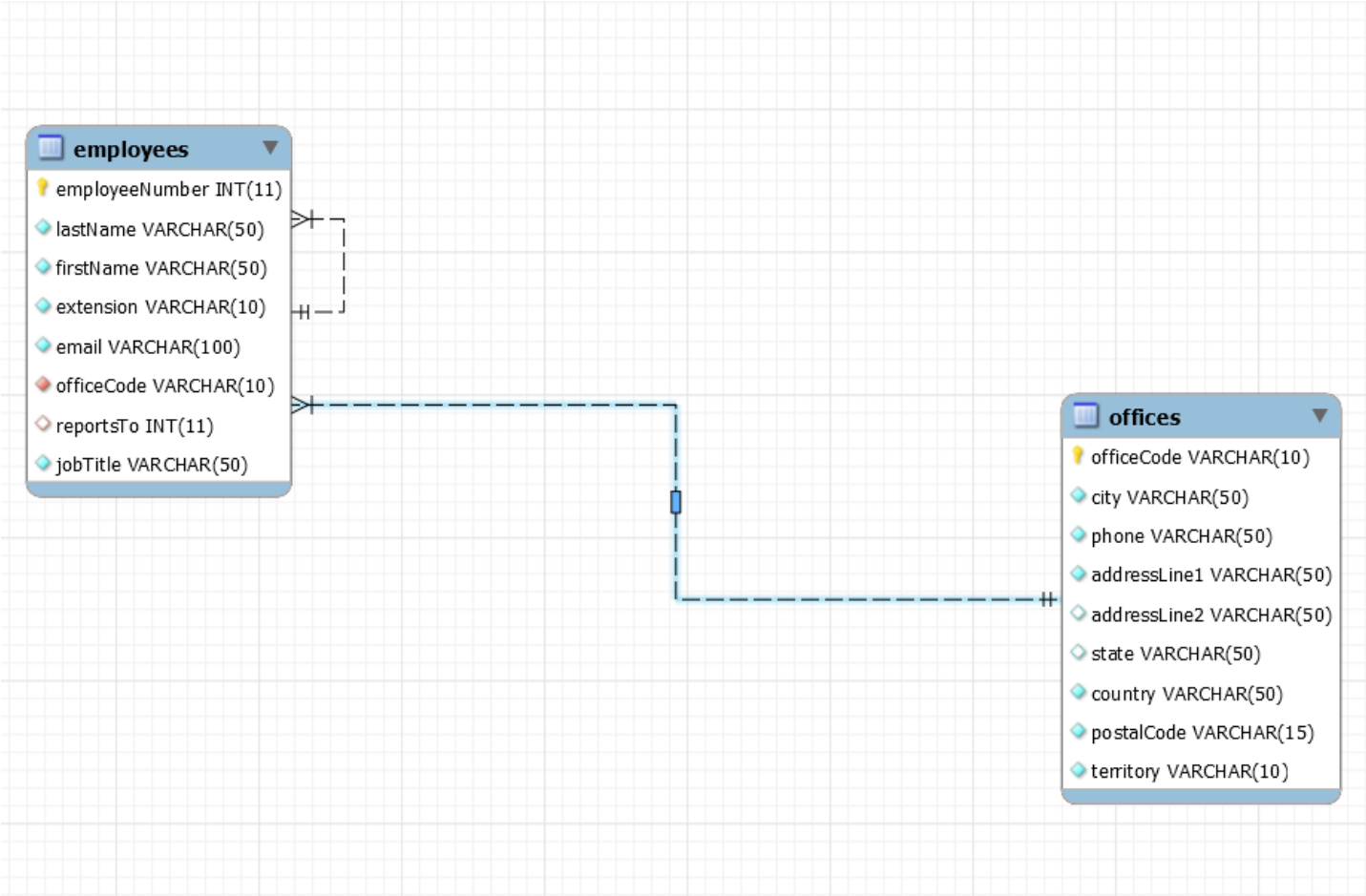


Lab 2: Reverse Engineer an ER Model





Lab 3: Forward Engineer an ER Model



Summary

- An RDBMS manages shared, structured data with integrity, security, and concurrency control.
- Design moves from conceptual (ER) to logical (tables, keys) to physical (indexes, partitioning).
- ER diagrams capture entities, relationships, and cardinality; an M:N relationship becomes a join table.
- Keys and constraints enforce integrity; normalization (1NF–3NF) removes redundancy and anomalies.
- Indexes speed up reads but add overhead to writes; administration keeps databases reliable in production.

Appendix: Contents

- A. Relational Model Reference: Codd's rules, data types, ER-to-table mapping
- B. Design and Normalization Reference: classification, goals, factors, ER steps
- C. Exercises and Case Studies: ER design and normalization practice



Appendix A: Relational Model Reference

Self-study reference

Codd's 12 Rules

- Thirteen rules (numbered 0 to 12) proposed by E.F. Codd for a DBMS to be considered relational
- Most RDBMS do not comply with all of the rules
- SQL is the standard language for the relational model

1. Rule 0: Foundation Rule
2. Rule 1: Information Rule
3. Rule 2: Guaranteed Access Rule
4. Rule 3: Systematic Treatment of NULL Values
5. Rule 4: Active Online Catalog
6. Rule 5: Comprehensive Data Sub-Language Rule
7. Rule 6: View Updating Rule
8. Rule 7: High-Level Insert, Update, and Delete
9. Rule 8: Physical Data Independence
10. Rule 9: Logical Data Independence
11. Rule 10: Integrity Independence
12. Rule 11: Distribution Independence
13. Rule 12: Non-Subversion Rule

Data Types

Category	Types
Numeric	INTEGER, INT, SMALLINT, MEDIUMINT, BIGINT
String	CHAR, VARCHAR, VARBINARY, BLOB, TEXT, ENUM, SET
Date and time	DATE, TIME, DATETIME, TIMESTAMP, YEAR
Logical	TINYINT



Appendix B: Design and Normalization Reference

Self-study reference

Database Classification

Dimension	Examples
Number of users	Single-user; multi-user (workgroup, enterprise)
Primary function	Operational (OLTP); analytical (OLAP)
Location	Centralized; distributed (partitioned, replicated)
Data type and structure	Relational; document, key-value, column, graph

Database Design Goals

- Meet business system requirements
- Data consistency and integrity
- Minimize data redundancy
- Optimize system performance
- Minimize database build and maintenance costs
- Maximize user and application flexibility

Database Design Factors

Factor	Example
Type of application	OLTP vs. OLAP
Data format and size	Financial systems
Data maintenance and support	Open source vs. proprietary
Hardware and software	OS, servers, etc.
Number of users and location	Centralized vs. distributed
Schedule and budget	Application or project constraints

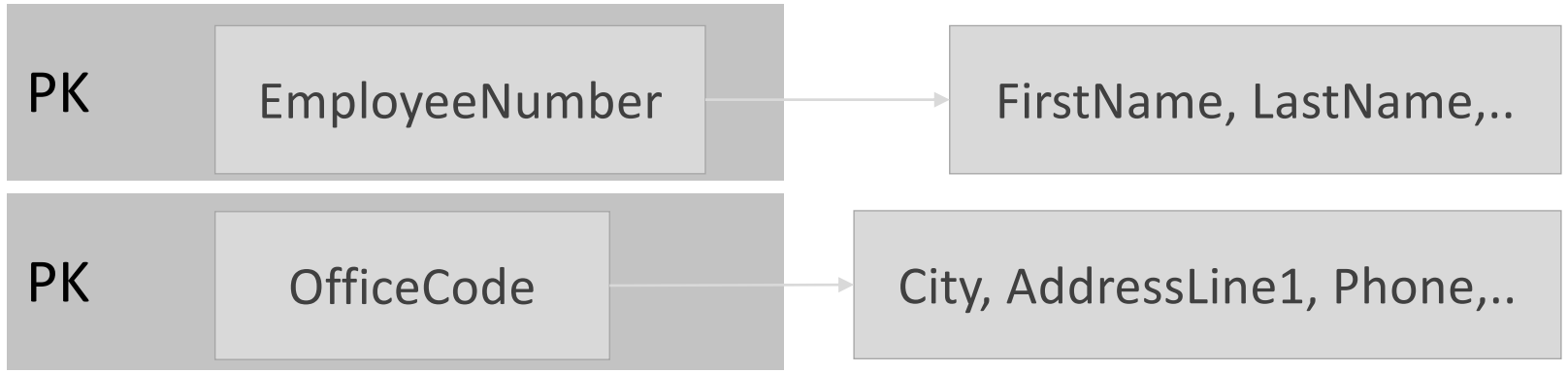
ER Diagram Steps

1. Identify and model entities and attributes
2. Identify unique identifiers for each entity
3. Analyze and model the relationships between the entities
4. Develop a complete ER model from requirements
5. Normalize the entities and relationships between them
6. Detailed EER modeling

Full Key Functional Dependency

- Primary key (PK) functionally determines the column of relation and no separate component of the PK partially determines the same column

e.g. Employees and Offices data in separate tables





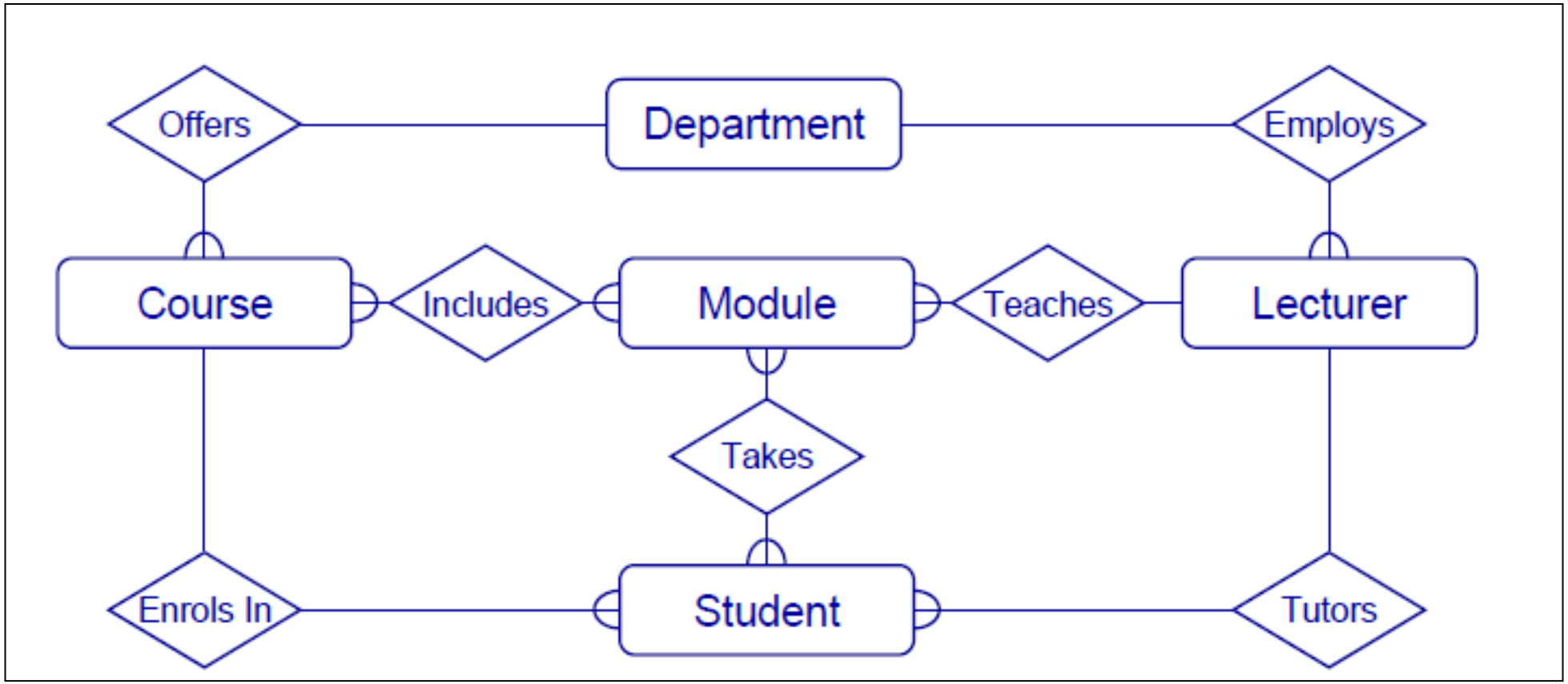
Appendix C: Exercises and Case Studies

Self-study reference

Practice 1: Identify Entities and Relationships

A university consists of a number of departments. Each department offers several courses. A number of modules make up each course. Students enroll in a particular course and take modules towards the completion of that course. Each module is taught by a lecturer from the appropriate department, and each lecturer tutors a group of students

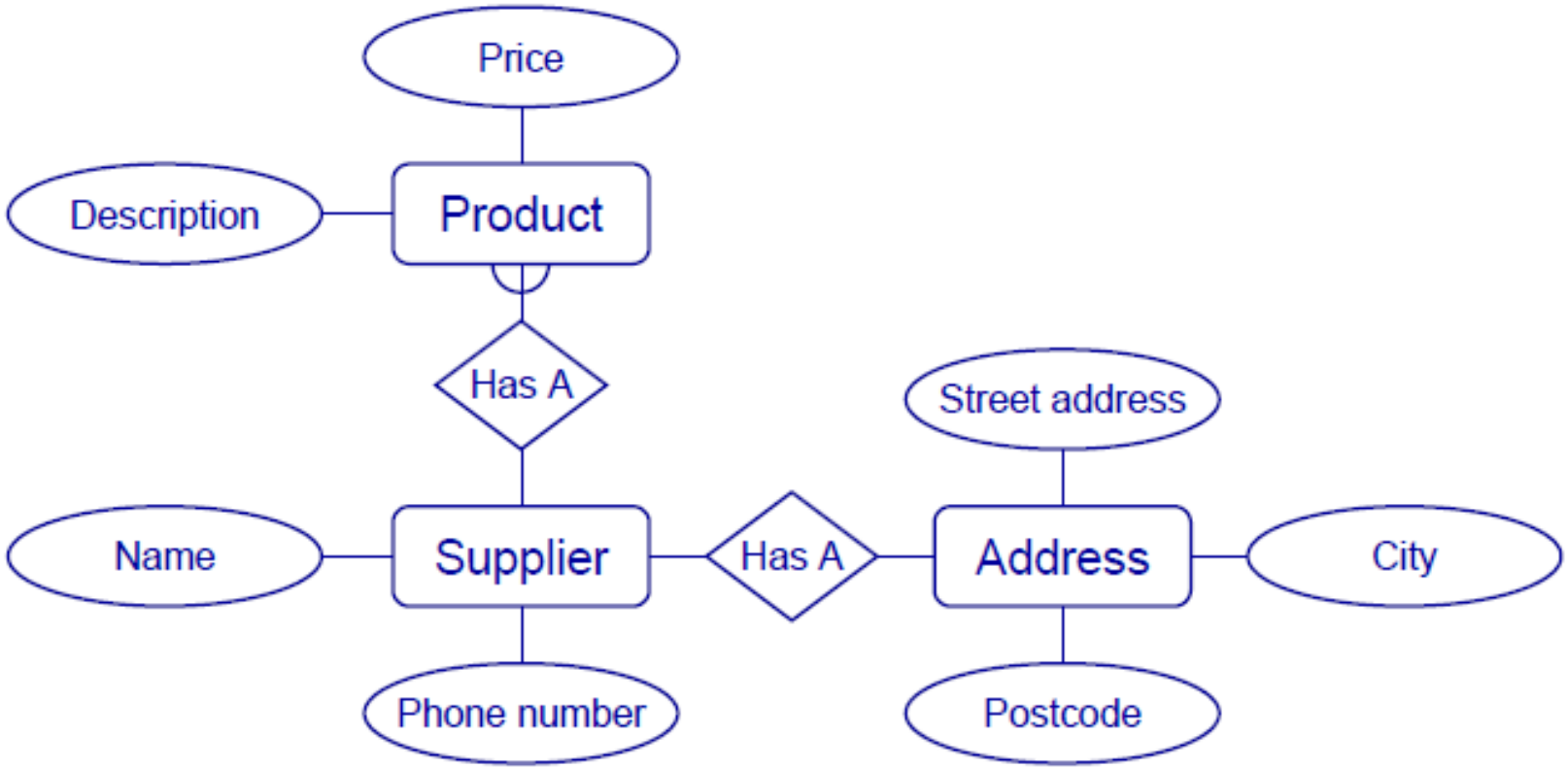
Practice 1: ER Diagram Solution



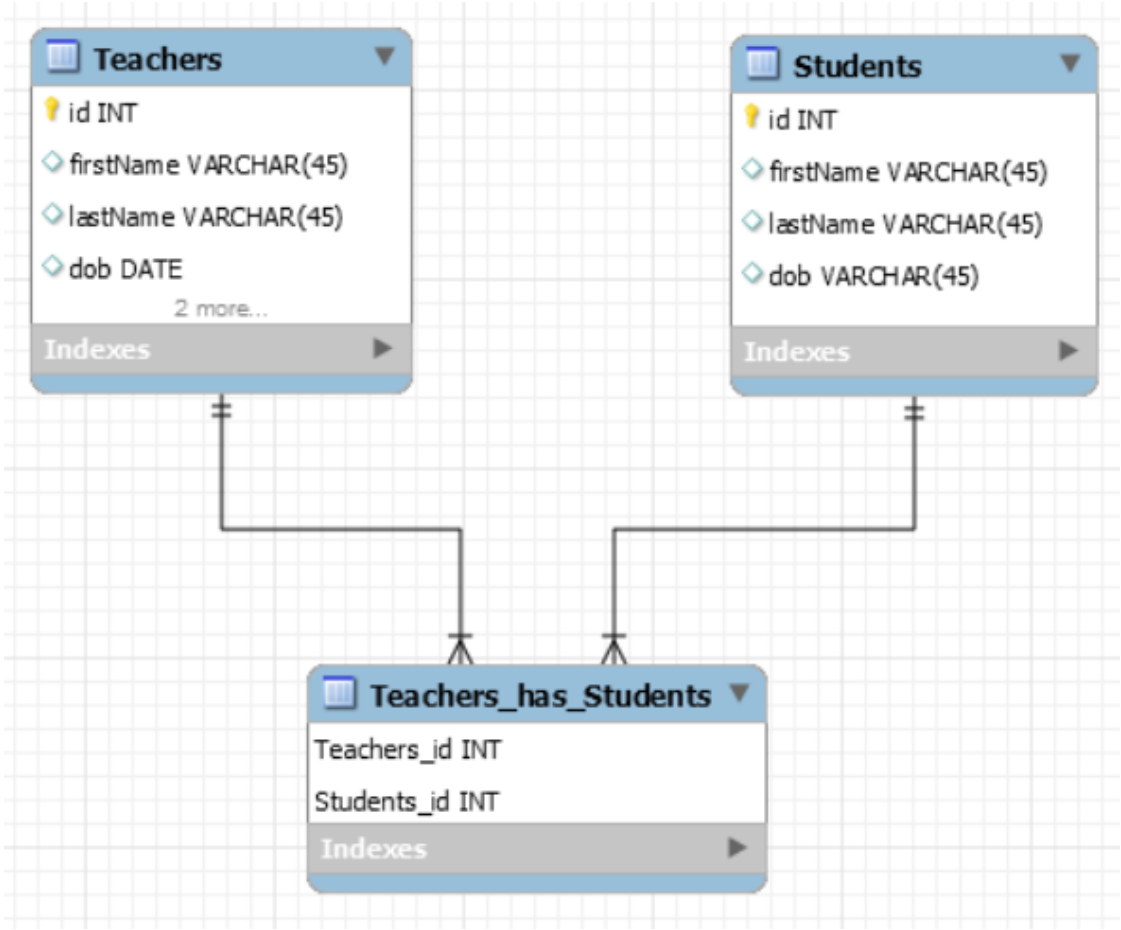
Practice 2: Identify Entities, Attributes, and Relationships

We want to represent information about products in a database. Each product has a description, a price and a supplier. Suppliers have addresses, phone numbers, and names. Each address is made up of a street address, a city, and a postcode

Practice 2: ER Diagram Solution



Practice 3: Create an ER Model and Forward Engineer



Practice 4: Normalize the Employees Table

Unnormalized Form (UNF):

employees

city	phone	address	employees
San Francisco	+1 650 219 4782	100 Market Street, Suite 300, San Francisco CA,USA, 94080	Murphy Diane, x5800, dmurphy@classicmodelcars.com, President Patterson Mary, x4611, mpatterso@classicmodelcars.com , VP Sales Firrelli Jeff, x9273, jfirrelli@classicmodelcars.com, VP Marketing Bow Anthony, x5428, abow@classicmodelcars.com, Sales Manager (NA) Jennings Leslie, x3291, ljennings@classicmodelcars.com, Sales Rep Thompson Leslie, x4065, lthompson@classicmodelcars.com, Sales Rep
Boston	+1 215 837 0825	1550 Court Place, Suite 102, Boston MA, USA, 2107	Firrelli Julie, x2173, jfirrelli@classicmodelcars.com, Sales Rep Patterson Steve, x4334, spatterson@classicmodelcars.com, Sales Rep
NYC	+1 212 555 3000	523 East 53rd Street, apt. 5A, New York City NY, USA, 10022	Tseng Foon Yue, x2248, ftseng@classicmodelcars.com, Sales Rep Vanauf George, x4102 gvanauf@classicmodelcars.com, Sales Rep
Paris	+33 14 723 4404	43 Rue Jouffroy D'abbans, Paris France, 75017	Bondur Gerard, x5408, gbondur@classicmodelcars.com, Sales Manager (EMEA) Bondur Loui, x6493, lbondur@classicmodelcars.com, Sales Rep Hernandez Gerard, x2028, ghernande@classicmodelcars.com, Sales Rep Castillo Pamela, x2759, pcastillo@classicmodelcars.com, Sales Rep Gerard Martin, x2312, mgerard@classicmodelcars.com, Sales Rep
Tokyo	+81 33 224 5000	4-1 Kioicho, Chiyoda-Ku, Tokyo Japan, 102-8578,	Nishi Mami, x101, mnishi@classicmodelcars.com,Sales Rep Kato Yoshimi, x102, ykato@classicmodelcars.com, Sales Rep
Sydney	+61 2 9264 2451	5-11 Wentworth Avenue, Floor #2, Sydney, Australia, NSW 2010	Patterson William, x4871, wpatterson@classicmodelcars.com, Sales Manager (APAC) Fixter Andy, x101, afixter@classicmodelcars.com, Sales Rep Marsh Peter, x102, pmarsh@classicmodelcars.com, Sales Rep King Tom, x103, tking@classicmodelcars.com, Sales Rep
London	+44 20 7877 204	25 Old Broad Street, Level 7, London UK, EC2N 1HN	Bott Larry, x2311, lbott@classicmodelcars.com, Sales Rep Jones Barry, x102, bjones@classicmodelcars.com, Sales Rep

Third Normal Form (3NF): 2NF + does not contain transitive functional dependencies

offices

officeCode (PK)	cityID(PK)	phone	addressLine1	addressLine2	postalCode
1	1001	+1 650 219 4782	100 Market Street	Suite 300	94080
2	1002	+1 215 837 0825	1550 Court Place	Suite 102	2107
3	1003	+1 212 555 3000	523 East 53rd Street	apt. 5A	10022
4	1004	+33 14 723 4404	43 Rue Jouffroy D'abba	NULL	75017
5	1005	+81 33 224 5000	4-1 Kioicho	NULL	102-8578
6	1006	+61 2 9264 2451	5-11 Wentworth Avenue	Floor #2	NSW 2010
7	1007	+44 20 7877 2041	25 Old Broad Street	Level 7	EC2N 1HN
8	1001	+1 650 476 4732	100 Fairfax Ave	Suite 1314	94110

cities

cityID(PK)	city	state	country
1001	San Francisc	CA	USA
1002	Boston	MA	USA
1003	NYC	NY	USA
1004	Paris	NULL	France
1005	Tokyo	Chiyoda-Ku	Japan
1006	Sydney	NULL	Australia
1007	London	NULL	UK

employees

employeeNumbe	lastName	firstName	extension	email	jobTitle	officeCode (FK)
1002	Murphy	Diane	x5800	dmurphy@classicmodelcars.com	President	1
1056	Patterson	Mary	x4611	mpatterso@classicmodelcars.com	VP Sales	1
1076	Firrelli	Jeff	x9273	jfirrelli@classicmodelcars.com	VP Marketing	1
1143	Bow	Anthony	x5428	abow@classicmodelcars.com	Sales Manager (NA)	1
1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	Sales Rep	1
1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.co	Sales Rep	1
1188	Firrelli	Julie	x2173	jfirrelli@classicmodelcars.com	Sales Rep	2
1216	Patterson	Steve	x4334	spatterson@classicmodelcars.co	Sales Rep	2
1286	Tseng	Foon Yue	x2248	ftseng@classicmodelcars.com	Sales Rep	3
1323	Vanauf	George	x4102	gvanauf@classicmodelcars.com	Sales Rep	3
1102	Bondur	Gerard	x5408	gbondur@classicmodelcars.com	Sales Manager (EME	4
1337	Bondur	Loui	x6493	lbondur@classicmodelcars.com	Sales Rep	4
1370	Hernandez	Gerard	x2028	ghernande@classicmodelcars.co	Sales Rep	4
1401	Castillo	Pamela	x2759	pcastillo@classicmodelcars.com	Sales Rep	4
1702	Gerard	Martin	x2312	mgerard@classicmodelcars.com	Sales Rep	4
1621	Nishi	Mami	x101	mnishi@classicmodelcars.com	Sales Rep	5
1625	Kato	Yoshimi	x102	ykato@classicmodelcars.com	Sales Rep	5
1088	Patterson	William	x4871	wpatterson@classicmodelcars.co	Sales Manager (APA	6
1611	Fixter	Andy	x101	afixter@classicmodelcars.com	Sales Rep	6
1612	Marsh	Peter	x102	pmarsh@classicmodelcars.com	Sales Rep	6
1619	King	Tom	x103	tking@classicmodelcars.com	Sales Rep	6
1501	Bott	Larry	x2311	lbott@classicmodelcars.com	Sales Rep	7
1504	Jones	Barry	x102	bjones@classicmodelcars.com	Sales Rep	7

Case Study A: Ad Campaign

AD CAMPAIGN MIX

AdCampaignID	AdCampaignName	StartDate	Duration	Campaign MgrID	Campaign MgrName	ModelID	Media	Range	BudgetPctg
111	SummerFun13	6.6.2013	12 days	CM100	Roberta	1	TV	Local	50%
111	SummerFun13	6.6.2013	12 days	CM100	Roberta	2	TV	National	50%
222	SummerZing13	6.8.2013	30 days	CM101	Sue	1	TV	Local	60%
222	SummerZing13	6.8.2013	30 days	CM101	Sue	3	Radio	Local	30%
222	SummerZing13	6.8.2013	30 days	CM101	Sue	5	Print	Local	10%
333	FallBall13	6.9.2013	12 days	CM102	John	3	Radio	Local	80%
333	FallBall13	6.9.2013	12 days	CM102	John	4	Radio	National	20%
444	AutmnStyle13	6.9.2013	5 days	CM103	Nancy	6	Print	National	100%
555	AutmnColors13	6.9.2013	3 days	CM100	Roberta	3	Radio	Local	100%

Case Study A Solution (1 of 2)

AD CAMPAIGN MIX

AdCampaignID	AdCampaignName	StartDate	Duration	Campaign MgrID	Campaign MgrName	ModelID	Media	Range	BudgetPctg
111	SummerFun13	6.6.2013	12 days	CM100	Roberta	1	TV	Local	50%
111	SummerFun13	6.6.2013	12 days	CM100	Roberta	2	TV	National	50%
222	SummerZing13	6.8.2013	**30 days**	CM101	Sue	1	TV	Local	60%
222	SummerZing13	6.8.2013	**30 days**	CM101	Sue	3	Radio	Local	30%
222	SummerZing13	6.8.2013	**30 days**	CM101	Sue	5	Print	Local	10%
333	FallBall13	6.9.2013	12 days	CM102	John	3	Radio	Local	80%
333	FallBall13	6.9.2013	12 days	CM102	John	4	Radio	National	20%
444	**AutmnStyle13**	**6.9.2013**	**5 days**	CM103	Nancy	6	Print	National	100%
555	AutmnColors13	6.9.2013	3 days	CM100	Roberta	3	Radio	Local	100%
????	????	????	????	????	????	**7**	**Internet**	**National**	????

Modification Anomaly Example:
To change the duration of the campaign 222 from 30 to 45 days, three records have to be modified

Deletion Anomaly Example:
Cannot delete campaign 444 without also deleting all the data about campaign manager CM103 and campaign mode 6

Insertion Anomaly Example:
Cannot insert new campaign mode 7 without inserting an actual campaign using the new mode 7

Case Study A Solution (2 of 2)

AD CAMPAIGN

AdCampaignID	AdCampaignName	StartDate	Duration	CampaignMgrID
111	SummerFun13	6.6.2013	12 days	CM100
222	SummerZing13	6.8.2013	30 days	CM101
333	FallBall13	6.9.2013	12 days	CM102
444	AutmnStyle13	6.9.2013	5 days	CM103
555	AutmnColors13	6.9.2013	3 days	CM100

CAMPAIGN MANAGER

CampaignMgrID	CampaignMgrName
CM100	Roberta
CM101	Sue
CM102	John
CM103	Nancy

MODE

ModeID	Media	Range
1	TV	Local
2	TV	National
3	Radio	Local
4	Radio	National
5	Print	Local
6	Print	National

AD CAMPAIGN MIX

AdCampaignID	ModeID	BudgetPctg
111	1	50%
111	2	50%
222	1	60%
222	3	30%
222	5	10%
333	3	80%
333	4	20%
444	6	100%
555	3	100%

Case Study B: Recruiting

RECRUITING

RecruiterID	RecruiterName	StatusID	Status	City	State	StatePopulation	CityPopulation	NoOfRecruits
R1	Katy	IF	Internal Full Time	Portland	ME	1,350,000	70,000	11
R1	Katy	IF	Internal Full Time	Grand Rapids	MI	9,900,000	190,000	20
R2	Abra	IP	Internal Part Time	Rockford	IL	12,900,000	340,000	17
R3	Jana	CN	Contractor	Spokane	WA	6,800,000	210,000	8
R3	Jana	CN	Contractor	Portland	OR	3,900,000	600,000	30
R3	Jana	CN	Contractor	Eugene	OR	3,900,000	360,000	20
R4	Maria	IF	Internal Full Time	Rockford	IL	12,900,000	340,000	14
R4	Maria	IF	Internal Full Time	Grand Rapids	MN	5,400,000	11,000	9
R5	Dan	CN	Contractor	Grand Rapids	MI	9,900,000	190,000	33

Case Study B Solution

RECRUITER

RecruiterID	RecruiterName	StatusID
R1	Katy	IF
R2	Abra	IP
R3	Jana	CN
R4	Maria	IF
R5	Dan	CN

STATUS

StatusID	Status
CN	Contractor
IF	Internal Full Time
IP	Internal Part Time

CITY

City	State	CityPopulation
Portland	ME	70,000
Grand Rapids	MI	190,000
Rockford	IL	340,000
Spokane	WA	210,000
Portland	OR	600,000
Eugene	OR	360,000
Grand Rapids	MN	11,000

STATE

State	StatePopulation
ME	1,350,000
MI	9,900,000
IL	12,900,000
WA	6,800,000
OR	3,900,000
MN	5,400,000

RECRUITING

RecruiterID	City	State	NoOfRecruits
R1	Portland	ME	11
R1	Grand Rapids	MI	20
R2	Rockford	IL	17
R3	Spokane	WA	8
R3	Portland	OR	30
R3	Eugene	OR	20
R4	Rockford	IL	14
R4	Grand Rapids	MN	9
R5	Grand Rapids	MI	33